beginning

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)

# Input data files are available in the read-only "../input/" directory
# For example, running this (by clicking run or pressing Shift+Enter) will list all files under the input directory

import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))
```

data loading

```
agg_encoded_20 = pd.read_excel("/content/drive/MyDrive/capstone/datasets/agg/agg_df_encoded_20.xlsx")
agg_encoded_40 = pd.read_excel("/content/drive/MyDrive/capstone/datasets/agg/agg_df_encoded_40.xlsx")

agg_knn_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/agg/agg_df_knn_context.xlsx")
agg_knn_no_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/agg/agg_df_knn_no_context.xlsx")

agg_mice_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/agg/agg_df_mice_context.xlsx")
agg_mice_no_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/agg/agg_df_mice_no_context.xlsx")

agg_rf_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/agg/agg_df_rf_context.xlsx")
agg_rf_no_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/agg/agg_df_rf_no_context.xlsx")




dbscan_encoded_20 = pd.read_excel("/content/drive/MyDrive/capstone/datasets/dbscan/dbscan_df_encoded_20.xlsx")
dbscan_encoded_40 = pd.read_excel("/content/drive/MyDrive/capstone/datasets/dbscan/dbscan_df_encoded_40.xlsx")

dbscan_knn_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/dbscan/dbscan_df_knn_context.xlsx")
dbscan_knn_no_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/dbscan/dbscan_df_knn_no_context.xlsx")

dbscan_mice_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/dbscan/dbscan_df_mice_context.xlsx")
dbscan_mice_no_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/dbscan/dbscan_df_mice_no_context.xlsx")

dbscan_rf_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/dbscan/dbscan_df_rf_context.xlsx")
dbscan_rf_no_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/dbscan/dbscan_df_rf_no_context.xlsx")
```

```
kmeans_encoded_20 = pd.read_excel("/content/drive/MyDrive/capstone/datasets/kmeans/kmeans_df_encoded_20.xlsx")
kmeans_encoded_40 = pd.read_excel("/content/drive/MyDrive/capstone/datasets/kmeans/kmeans_df_encoded_40.xlsx")

kmeans_knn_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/kmeans/kmeans_df_knn_context.xlsx")
kmeans_knn_no_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/kmeans/kmeans_df_knn_no_context.xlsx")

kmeans_mice_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/kmeans/kmeans_df_mice_context.xlsx")
kmeans_mice_no_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/kmeans/kmeans_df_mice_no_context.xlsx")

kmeans_rf_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/kmeans/kmeans_df_rf_context.xlsx")
kmeans_rf_no_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/kmeans/kmeans_df_rf_no_context.xlsx")



spc_encoded_20 = pd.read_excel("/content/drive/MyDrive/capstone/datasets/spc/spc_df_encoded_20.xlsx")
spc_encoded_40 = pd.read_excel("/content/drive/MyDrive/capstone/datasets/spc/spc_df_encoded_20.xlsx")

spc_knn_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/spc/spc_df_knn_context.xlsx")
spc_knn_no_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/spc/spc_df_knn_no_context.xlsx")

spc_mice_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/spc/spc_df_mice_context.xlsx")
spc_mice_no_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/spc/spc_df_mice_no_context.xlsx")

spc_rf_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/spc/spc_df_rf_context.xlsx")
spc_rf_no_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/spc/spc_df_rf_no_context.xlsx")
```

dataframe grouping

```
agg_dfs = [
    # Agglomerative
    (agg_encoded_20, "agg_encoded_20"),
    (agg_encoded_40, "agg_encoded_40"),
    (agg_knn_context, "agg_knn_context"),
    (agg_knn_no_context, "agg_knn_no_context"),
    (agg_mice_context, "agg_mice_context"),
    (agg_mice_no_context, "agg_mice_no_context"),
    (agg_rf_context, "agg_rf_context"),
    (agg_rf_no_context, "agg_rf_no_context"),
]

dbscan_dfs = [
    # DBSCAN
    (dbscan_encoded_20, "dbscan_encoded_20"),
    (dbscan_encoded_40, "dbscan_encoded_40"),
    (dbscan_knn_context, "dbscan_knn_context"),
    (dbscan_knn_no_context, "dbscan_knn_no_context"),
    (dbscan_mice_context, "dbscan_mice_context"),
    (dbscan_mice_no_context, "dbscan_mice_no_context"),
    (dbscan_rf_context, "dbscan_rf_context"),
    (dbscan_rf_no_context, "dbscan_rf_no_context"),
]
```

```
kmeans_dfs = [
    # KMeans
    (kmeans_encoded_20, "kmeans_encoded_20"),
    (kmeans_encoded_40, "kmeans_encoded_40"),
    (kmeans_knn_context, "kmeans_knn_context"),
    (kmeans_knn_no_context, "kmeans_knn_no_context"),
    (kmeans_mice_context, "kmeans_mice_context"),
    (kmeans_mice_no_context, "kmeans_mice_no_context"),
    (kmeans_rf_context, "kmeans_rf_context"),
    (kmeans_rf_no_context, "kmeans_rf_no_context")
]

spc_dfs = [
    # spc
    (spc_encoded_20, "spc_encoded_20"),
    (spc_encoded_40, "spc_encoded_40"),
    (spc_knn_context, "spc_knn_context"),
    (spc_knn_no_context, "spc_knn_no_context"),
    (spc_mice_context, "spc_mice_context"),
    (spc_mice_no_context, "spc_mice_no_context"),
    (spc_rf_context, "spc_rf_context"),
    (spc_rf_no_context, "spc_rf_no_context")
]


# Combine all three into one list
all_dfs = agg_dfs + dbscan_dfs + kmeans_dfs + spc_dfs


# extra groups for cross comparison
no_context_dfs = [
    (agg_knn_no_context, "agg_knn_no_context"),
    (agg_mice_no_context, "agg_mice_no_context"),
    (agg_rf_no_context, "agg_rf_no_context"),
    (spc_knn_no_context, "spc_knn_no_context"),
    (dbscan_knn_no_context, "dbscan_knn_no_context"),
    (dbscan_mice_no_context, "dbscan_mice_no_context"),
    (spc_mice_no_context, "spc_mice_no_context"),
    (dbscan_rf_no_context, "dbscan_rf_no_context"),
    (kmeans_knn_no_context, "kmeans_knn_no_context"),
    (kmeans_mice_no_context, "kmeans_mice_no_context"),
    (kmeans_rf_no_context, "kmeans_rf_no_context"),
    (spc_rf_no_context, "spc_rf_no_context"),
]

context_dfs = [
    (agg_knn_context, "agg_knn_context"),
    (agg_mice_context, "agg_mice_context"),
    (agg_rf_context, "agg_rf_context"),
    (dbscan_knn_context, "dbscan_knn_context"),
    (dbscan_mice_context, "dbscan_mice_context"),
    (dbscan_rf_context, "dbscan_rf_context"),
    (kmeans_knn_context, "kmeans_knn_context"),
    (kmeans_mice_context, "kmeans_mice_context"),
    (kmeans_rf_context, "kmeans_rf_context"),
    (spc_knn_context, "spc_knn_context"),
```

```
    (spc_mice_context, "spc_mice_context"),
    (spc_rf_context, "spc_rf_context"),
]

no_imputed_dfs = [
    (agg_encoded_20, "agg_encoded_20"),
    (agg_encoded_40, "agg_encoded_40"),
    (dbscan_encoded_20, "dbscan_encoded_20"),
    (dbscan_encoded_40, "dbscan_encoded_40"),
    (kmeans_encoded_20, "kmeans_encoded_20"),
    (kmeans_encoded_40, "kmeans_encoded_40"),
    (spc_encoded_20, "spc_encoded_20"),
    (spc_encoded_40, "spc_encoded_40"),
]

mice_dfs = [
    (agg_mice_context, "agg_mice_context"),
    (agg_mice_no_context, "agg_mice_no_context"),
    (dbscan_mice_context, "dbscan_mice_context"),
    (dbscan_mice_no_context, "dbscan_mice_no_context"),
    (kmeans_mice_context, "kmeans_mice_context"),
    (kmeans_mice_no_context, "kmeans_mice_no_context"),
    (spc_mice_context, "spc_mice_context"),
    (spc_mice_no_context, "spc_mice_no_context"),
]

rf_dfs = [
    (agg_rf_context, "agg_rf_context"),
    (agg_rf_no_context, "agg_rf_no_context"),
    (dbscan_rf_context, "dbscan_rf_context"),
    (dbscan_rf_no_context, "dbscan_rf_no_context"),
    (kmeans_rf_context, "kmeans_rf_context"),
    (kmeans_rf_no_context, "kmeans_rf_no_context"),
    (spc_rf_context, "spc_rf_context"),
    (spc_rf_no_context, "spc_rf_no_context")
]

knn_dfs = [
    (agg_knn_context, "agg_knn_context"),
    (agg_knn_no_context, "agg_knn_no_context"),
    (dbscan_knn_context, "dbscan_knn_context"),
    (dbscan_knn_no_context, "dbscan_knn_no_context"),
    (kmeans_knn_context, "kmeans_knn_context"),
    (kmeans_knn_no_context, "kmeans_knn_no_context"),
    (spc_knn_context, "spc_knn_context"),
    (spc_knn_no_context, "spc_knn_no_context"),
]
```

general evaluation function

```
from sklearn.metrics import silhouette_score, calinski_harabasz_score, davies_bouldin_score
from scipy.spatial.distance import cdist
import numpy as np
```

```
def evaluate_clusters(
    df,
    df_name,
    sil_metric="euclidean",  # match your clustering distance
    sample_size=None,
    random_state=42,
    return_dict=False
):
    """
    General-purpose clustering evaluation function.
    Works for Agglomerative, DBSCAN, KMeans, Spectral, GMM, etc.
    """

    rng = np.random.RandomState(random_state)

    label_col = df_name.split("_")[0] + "_cluster"

    X = df.drop(columns=[label_col]).values
    labels = df[label_col].values

    # Optionally ignore a label (e.g., DBSCAN noise = -1)
    if df_name.split("_")[0] == 'dbscan':
        mask = labels != -1
        X, labels = X[mask], labels[mask]

    # Optional subsample for speed (applies to all metrics)
    if sample_size is not None and X.shape[0] > sample_size:
        idx = rng.choice(X.shape[0], size=sample_size, replace=False)
        X, labels = X[idx], labels[idx]

    unique_clusters, counts = np.unique(labels, return_counts=True)
    n_clusters = len(unique_clusters)

    if n_clusters <= 1:
        print("⚠ Only one cluster found. Metrics cannot be computed robustly.")
        return {} if return_dict else None

    results = {}

    # Core indices
    try:
        results["Silhouette Score"] = {
            "score": silhouette_score(X, labels, metric=sil_metric),
            "range": "[-1, 1]",
            "better": "higher"
        }
    except Exception:
        results["Silhouette Score"] = {"score": np.nan, "range": "[-1, 1]", "better": "higher"}

    try:
        results["Calinski–Harabasz Index"] = {
            "score": calinski_harabasz_score(X, labels),
            "range": "[0, ∞)",
            "better": "higher"
        }
```

```
except Exception:
    results["Calinski–Harabasz Index"] = {"score": np.nan, "range": "[0, ∞)", "better": "higher"}

try:
    results["Davies–Bouldin Index"] = {
        "score": davies_bouldin_score(X, labels),
        "range": "[0, ∞)",
        "better": "lower"
    }
except Exception:
    results["Davies–Bouldin Index"] = {"score": np.nan, "range": "[0, ∞)", "better": "lower"}

# Dunn Index
def dunn_index(X, labels):
    eps = 1e-12
    uniq = np.unique(labels)
    # Intra-cluster diameters
    intra = []
    for k in uniq:
        pts = X[labels == k]
        if pts.shape[0] >= 2:
            d = cdist(pts, pts)
            intra.append(np.max(d))
        else:
            intra.append(0.0)  # singleton cluster
    denom = max(max(intra), eps)

    # Inter-cluster minimum distance
    inter_min = np.inf
    for i in range(len(uniq)):
        for j in range(i + 1, len(uniq)):
            Xi = X[labels == uniq[i]]
            Xj = X[labels == uniq[j]]
            d = cdist(Xi, Xj)
            inter_min = min(inter_min, np.min(d))
    if not np.isfinite(inter_min):
        return np.nan
    return inter_min / denom

# Xie–Beni Index
def xie_beni_index(X, labels):
    eps = 1e-12
    uniq = np.unique(labels)
    centroids = np.array([X[labels == k].mean(axis=0) for k in uniq])
    # Compactness
    compact = 0.0
    for i, k in enumerate(uniq):
        pts = X[labels == k]
        diff = pts - centroids[i]
        compact += np.sum(diff * diff)
    # Min centroid separation
    min_csep = np.inf
    for i in range(len(centroids)):
        for j in range(i + 1, len(centroids)):
            d = np.sum((centroids[i] - centroids[j]) ** 2)
            if d < min_csep:
```

```
                    min_csep = d
            if not np.isfinite(min_csep) or min_csep <= eps:
                return np.nan
            return compact / (X.shape[0] * min_csep)

        try:
            results["Dunn Index"] = {
                "score": dunn_index(X, labels),
                "range": "[0, ∞)",
                "better": "higher"
            }
        except Exception:
            results["Dunn Index"] = {"score": np.nan, "range": "[0, ∞)", "better": "higher"}

        try:
            results["Xie–Beni Index"] = {
                "score": xie_beni_index(X, labels),
                "range": "[0, ∞)",
                "better": "lower"
            }
        except Exception:
            results["Xie–Beni Index"] = {"score": np.nan, "range": "[0, ∞)", "better": "lower"}

        # Pretty print
        print("\n📊 Clustering Evaluation Results:")
        for metric, info in results.items():
            score = info['score']
            score_str = f"{score:.4f}" if isinstance(score, (int, float)) and np.isfinite(score) else "NaN"
            print(f"{metric}: {score_str} --------- Range: {info['range']} ----------- {info['better'].capitalize()} is better")

        # Final cluster summary in one line
        cluster_summary = ", ".join([f"{c}:{cnt}" for c, cnt in zip(unique_clusters, counts)])
        print(f"📦 Total Clusters: {n_clusters} | Cluster Sizes: {cluster_summary}")

        return results if return_dict else None
```

```
# --- Evaluate KMeans ---
for df, name in all_dfs:
    print(f"\nEvaluating {name} (KMeans)")
    evaluate_clusters(
        df,
        df_name=name,
        sil_metric="euclidean",
        sample_size=None,
        return_dict=False
    )
```

```
📊 Clustering Evaluation Results:
Silhouette Score: 0.7954 --------- Range: [-1, 1] ----------- Higher is better
Calinski–Harabasz Index: 9875.8666 --------- Range: [0, ∞) ----------- Higher is better
Davies–Bouldin Index: 0.2486 --------- Range: [0, ∞) ----------- Lower is better
Dunn Index: 0.0257 --------- Range: [0, ∞) ----------- Higher is better
Xie–Beni Index: 0.0315 --------- Range: [0, ∞) ----------- Lower is better
📦 Total Clusters: 2 | Cluster Sizes: 0:422, 1:1195

Evaluating spc_mice_context (KMeans)

📊 Clustering Evaluation Results:
Silhouette Score: 0.6708 --------- Range: [-1, 1] ----------- Higher is better
Calinski–Harabasz Index: 5302.2638 --------- Range: [0, ∞) ----------- Higher is better
Davies–Bouldin Index: 0.4260 --------- Range: [0, ∞) ----------- Lower is better
Dunn Index: 0.0018 --------- Range: [0, ∞) ----------- Higher is better
Xie–Beni Index: 0.0759 --------- Range: [0, ∞) ----------- Lower is better
📦 Total Clusters: 2 | Cluster Sizes: 0:795, 1:818

Evaluating spc_mice_no_context (KMeans)

📊 Clustering Evaluation Results:
Silhouette Score: 0.7878 --------- Range: [-1, 1] ----------- Higher is better
Calinski–Harabasz Index: 12376.4397 --------- Range: [0, ∞) ----------- Higher is better
Davies–Bouldin Index: 0.2671 --------- Range: [0, ∞) ----------- Lower is better
Dunn Index: 0.0212 --------- Range: [0, ∞) ----------- Higher is better
Xie–Beni Index: 0.0322 --------- Range: [0, ∞) ----------- Lower is better
📦 Total Clusters: 2 | Cluster Sizes: 0:883, 1:730

Evaluating spc_rf_context (KMeans)

📊 Clustering Evaluation Results:
Silhouette Score: 0.7548 --------- Range: [-1, 1] ----------- Higher is better
Calinski–Harabasz Index: 9686.6116 --------- Range: [0, ∞) ----------- Higher is better
Davies–Bouldin Index: 0.3716 --------- Range: [0, ∞) ----------- Lower is better
Dunn Index: 0.9821 --------- Range: [0, ∞) ----------- Higher is better
Xie–Beni Index: 0.0415 --------- Range: [0, ∞) ----------- Lower is better
📦 Total Clusters: 2 | Cluster Sizes: 0:858, 1:759

Evaluating spc_rf_no_context (KMeans)

📊 Clustering Evaluation Results:
Silhouette Score: 0.7564 --------- Range: [-1, 1] ----------- Higher is better
Calinski–Harabasz Index: 9792.4979 --------- Range: [0, ∞) ----------- Higher is better
Davies–Bouldin Index: 0.3693 --------- Range: [0, ∞) ----------- Lower is better
Dunn Index: 0.9810 --------- Range: [0, ∞) ----------- Higher is better
Xie–Beni Index: 0.0411 --------- Range: [0, ∞) ----------- Lower is better
📦 Total Clusters: 2 | Cluster Sizes: 0:858, 1:759
```

combined evaluation function

```
import numpy as np
from collections import Counter
from scipy.spatial.distance import cdist
from sklearn.metrics import silhouette_score, calinski_harabasz_score, davies_bouldin_score

# ==============================
# Config: metric order & weights
# ------------------------------
```

```
# ==============================
METRIC_ORDER = ["balance", "silhouette", "calinski_harabasz", "dunn", "davies_bouldin", "xie_beni"]
METRIC_WEIGHTS = {
    "balance": 0.35,
    "silhouette": 0.25,
    "calinski_harabasz": 0.20,
    "dunn": 0.10,
    "davies_bouldin": 0.05,
    "xie_beni": 0.05,
}

# ==================================
# Utilities: detect label column/name
# ==================================
def get_algo_and_label(df_name: str):
    if df_name.startswith("dbscan"):
        return "dbscan", "dbscan_cluster"
    if df_name.startswith("kmeans"):
        return "kmeans", "kmeans_cluster"
    if df_name.startswith("agg"):
        return "agg", "agg_cluster"
    if df_name.startswith("spc"):
        return "spc", "spc_cluster"
    raise ValueError(f"Unknown algorithm for {df_name}")

# ========================
# Metric helper calculations
# ========================
def _safe_min_pairwise_dist(M):
    if M.size == 0:
        return np.nan
    mask = ~np.eye(M.shape[0], dtype=bool)
    vals = M[mask]
    vals = vals[np.isfinite(vals)]
    if vals.size == 0:
        return np.nan
    return np.min(vals)

def _dunn_index(X, labels):
    uniq = np.unique(labels)
    if uniq.size < 2:
        return np.nan
    intramax = 0.0
    for k in uniq:
        pts = X[labels == k]
        if pts.shape[0] < 2:
            continue
        D = cdist(pts, pts)
        intramax = max(intramax, np.max(D))
    if not np.isfinite(intramax) or intramax <= 0:
        return np.nan
    inter_min = np.inf
    for i in range(uniq.size):
        for j in range(i+1, uniq.size):
            Xi = X[labels == uniq[i]]
            Xj = X[labels == uniq[j]]
            if Xi.size == 0 or Xj.size == 0:
```

```
                continue
            D = cdist(Xi, Xj)
            m = np.min(D) if D.size else np.inf
            inter_min = min(inter_min, m)
    if not np.isfinite(inter_min) or inter_min <= 0:
        return np.nan
    return inter_min / intramax

def _xie_beni_inv(X, labels):
    uniq = np.unique(labels)
    if uniq.size < 2:
        return np.nan
    cents = np.array([X[labels == k].mean(axis=0) for k in uniq])
    compact = 0.0
    for i, k in enumerate(uniq):
        pts = X[labels == k]
        if pts.size == 0:
            continue
        diff = pts - cents[i]
        compact += np.sum(diff * diff)
    C = cdist(cents, cents)
    min_csep = _safe_min_pairwise_dist(C)
    if not np.isfinite(min_csep) or min_csep <= 0:
        return np.nan
    xb = compact / (X.shape[0] * (min_csep ** 2))
    return 1.0 / (1.0 + xb)

def _cluster_balance(labels, k_penalty=True):
    counts = np.array(list(Counter(labels).values()), dtype=float)
    K = counts.size
    if K < 2:
        return 0.0
    props = counts / counts.sum()
    base = 1.0 - np.std(props)
    base = max(0.0, float(base))
    if not k_penalty:
        return base
    penalty = 1.0 / np.sqrt(K)
    return base * penalty

# ==========================================
# Module 1: Compute metrics for a single frame
# ==========================================
def compute_all_metrics_for_df(df, df_name):
    algo, label_col = get_algo_and_label(df_name)
    labels_all = df[label_col].values
    X_all = df.drop(columns=[label_col]).values

    balance = _cluster_balance(labels_all, k_penalty=True)

    if algo == "dbscan":
        mask = labels_all != -1
        if mask.sum() < 2 or len(np.unique(labels_all[mask])) < 2:
            return {
                "balance": balance,
                "silhouette": np.nan,
                "calinski_harabasz": np.nan,
```

```
                "davies_bouldin": np.nan,
                "dunn": np.nan,
                "xie_beni": np.nan,
            }
        X = X_all[mask]
        labels = labels_all[mask]
    else:
        X = X_all
        labels = labels_all
        if len(np.unique(labels)) < 2:
            return {
                "balance": balance,
                "silhouette": np.nan,
                "calinski_harabasz": np.nan,
                "davies_bouldin": np.nan,
                "dunn": np.nan,
                "xie_beni": np.nan,
            }

    def _try(fn):
        try:
            v = fn()
            return float(v) if np.isfinite(v) else np.nan
        except Exception:
            return np.nan

    sil = _try(lambda: silhouette_score(X, labels))
    ch = _try(lambda: calinski_harabasz_score(X, labels))
    db_inv = _try(lambda: 1.0 / (1.0 + davies_bouldin_score(X, labels)))
    dunn = _try(lambda: _dunn_index(X, labels))
    xb_inv = _try(lambda: _xie_beni_inv(X, labels))

    return {
        "balance": balance,
        "silhouette": sil,
        "calinski_harabasz": ch,
        "davies_bouldin": db_inv,
        "dunn": dunn,
        "xie_beni": xb_inv,
    }

# ========================================
# Module 2: Evaluate all and store results
# ========================================
all_evaluation_results = {}
metric_lists = {m: [] for m in METRIC_ORDER}
composite_list = []

def evaluate_all_datasets(all_dfs):
    for df, df_name in all_dfs:
        metrics = compute_all_metrics_for_df(df, df_name)
        all_evaluation_results[df_name] = metrics
        for m in METRIC_ORDER:
            metric_lists[m].append(metrics[m])

# ==================================
# Module 3: Ranking & final composite
```

```
# ==================================
def _sort_key_nan_last(val):
    return (-np.inf if np.isnan(val) else val)

def rank_per_metric(results_dict, metric):
    items = []
    for name, mdict in results_dict.items():
        v = mdict.get(metric, np.nan)
        items.append((name, v))
    items.sort(key=lambda x: _sort_key_nan_last(x[1]), reverse=True)
    return items

def composite_ranking(results_dict):
    per_metric_sorted = {m: rank_per_metric(results_dict, m) for m in METRIC_ORDER}
    n = len(results_dict)
    scores = {name: 0.0 for name in results_dict.keys()}
    for metric in METRIC_ORDER:
        weight = METRIC_WEIGHTS[metric]
        ranked_names = [name for name, _ in per_metric_sorted[metric]]
        for pos, name in enumerate(ranked_names):
            pts = (n - pos)
            val = results_dict[name].get(metric, np.nan)
            if np.isnan(val):
                pts = 0
            scores[name] += weight * pts
    final = sorted(scores.items(), key=lambda x: x[1], reverse=True)
    for name, score in final:
        composite_list.append(score)
    return per_metric_sorted, final

# ===========================
# Module 4: Pretty-printing
# ===========================
def print_rankings(per_metric_sorted, final_ranking, results_dict):
    print("\n📊 Ranked list for each metric:\n")
    for metric in METRIC_ORDER:
        print(f"--- {metric.upper()} ---")
        for name, _ in per_metric_sorted[metric]:
            val = results_dict[name][metric]
            if np.isnan(val):
                print(f"{name}: NaN")
            else:
                print(f"{name}: {val:.4f}")
        print()

    print("✅ Final Composite Ranking:\n")
    for i, (name, score) in enumerate(final_ranking, start=1):
        print(f"{i:>2}. {name}: Composite Score = {score:.4f}")

# =================
# Module 5: Run all
# =================
# Expect: all_dfs = [(df, "df_name"), ...] to be defined in your session.
evaluate_all_datasets(all_dfs)
per_metric_sorted, final_ranking = composite_ranking(all_evaluation_results)
```

```
# print_rankings(per_metric_sorted, final_ranking, all_evaluation_results)

# ✅ metric_lists contains per-metric values for all datasets
# ✅ composite_list contains the final composite scores
```

```
silhouette_ranking = per_metric_sorted["silhouette"]
final_ranking
```

```
[('agg_rf_no_context', 27.650000000000002),
 ('kmeans_mice_no_context', 26.949999999999996),
 ('agg_rf_context', 26.700000000000003),
 ('spc_rf_no_context', 25.95),
 ('dbscan_rf_context', 25.5),
 ('spc_mice_no_context', 25.449999999999996),
 ('spc_knn_no_context', 24.8),
 ('spc_rf_context', 24.8),
 ('spc_knn_context', 24.7),
 ('kmeans_rf_no_context', 23.65),
 ('kmeans_knn_no_context', 22.750000000000004),
 ('kmeans_knn_context', 22.450000000000003),
 ('kmeans_rf_context', 22.45),
 ('spc_mice_context', 20.850000000000005),
 ('dbscan_mice_context', 18.950000000000003),
 ('kmeans_mice_context', 18.699999999999996),
 ('dbscan_rf_no_context', 12.450000000000001),
 ('spc_encoded_20', 11.899999999999999),
 ('dbscan_encoded_20', 11.4),
 ('spc_encoded_40', 10.9),
 ('agg_encoded_40', 10.8),
 ('dbscan_encoded_40', 10.399999999999999),
 ('agg_mice_no_context', 9.85),
 ('agg_knn_context', 9.65),
 ('agg_knn_no_context', 9.35),
 ('kmeans_encoded_40', 9.35),
 ('dbscan_mice_no_context', 8.850000000000001),
 ('agg_mice_context', 8.45),
 ('kmeans_encoded_20', 7.3999999999999995),
 ('agg_encoded_20', 5.750000000000001),
 ('dbscan_knn_no_context', 4.75),
 ('dbscan_knn_context', 4.45)]
```

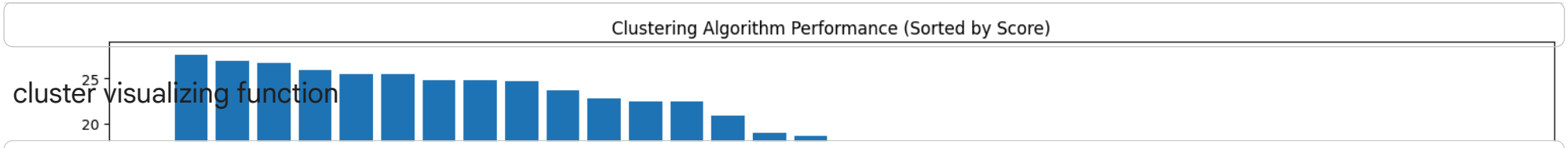
sorting dfs based on combined values

```
# Create a dictionary mapping dataframe names to their composite scores
# final_ranking is a list of (name, score) tuples, sorted descending
score_map = dict(final_ranking)

# Sort all_dfs based on the composite scores
all_dfs_sorted = sorted(
    all_dfs,
    key=lambda x: score_map.get(x[1], -np.inf),  # x[1] is df_name, use -inf for missing names
    reverse=True  # highest combined score first
)
```

bar plot to show combined value

```
import matplotlib.pyplot as plt

def plot_clustering_results(results_dict):
    """
    Generates a bar chart from clustering evaluation results provided as a dictionary.

    Args:
        results_dict (dict): A dictionary where keys are dataframe names and values are the scores.
    """
    # Extract names and scores from the dictionary
    names = list(results_dict.keys())
    scores = list(results_dict.values())

    # Create the bar chart
    plt.figure(figsize=(15, 5))
    plt.bar(names, scores)
    plt.xticks(rotation=90)
    plt.ylabel("Score")
    plt.title("Clustering Algorithm Performance (Sorted by Score)")
    plt.tight_layout()
    plt.show()

# Example usage (you can call this function with your results_dict)
# plot_clustering_results(my_results_dict)
```

```
# Example usage (you can call this function with your sorted_results)
plot_clustering_results(score_map)
```



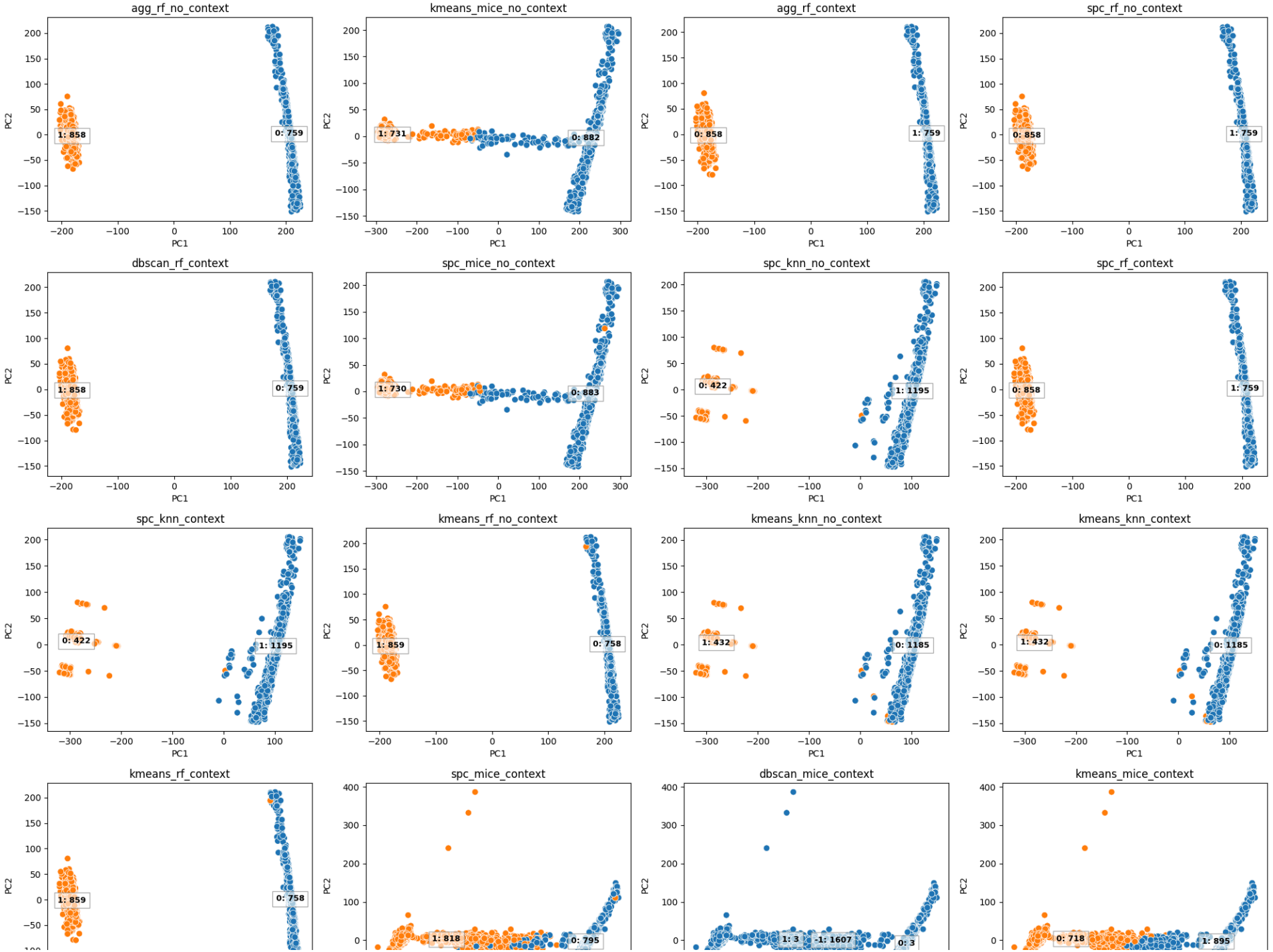
cluster visualizing function

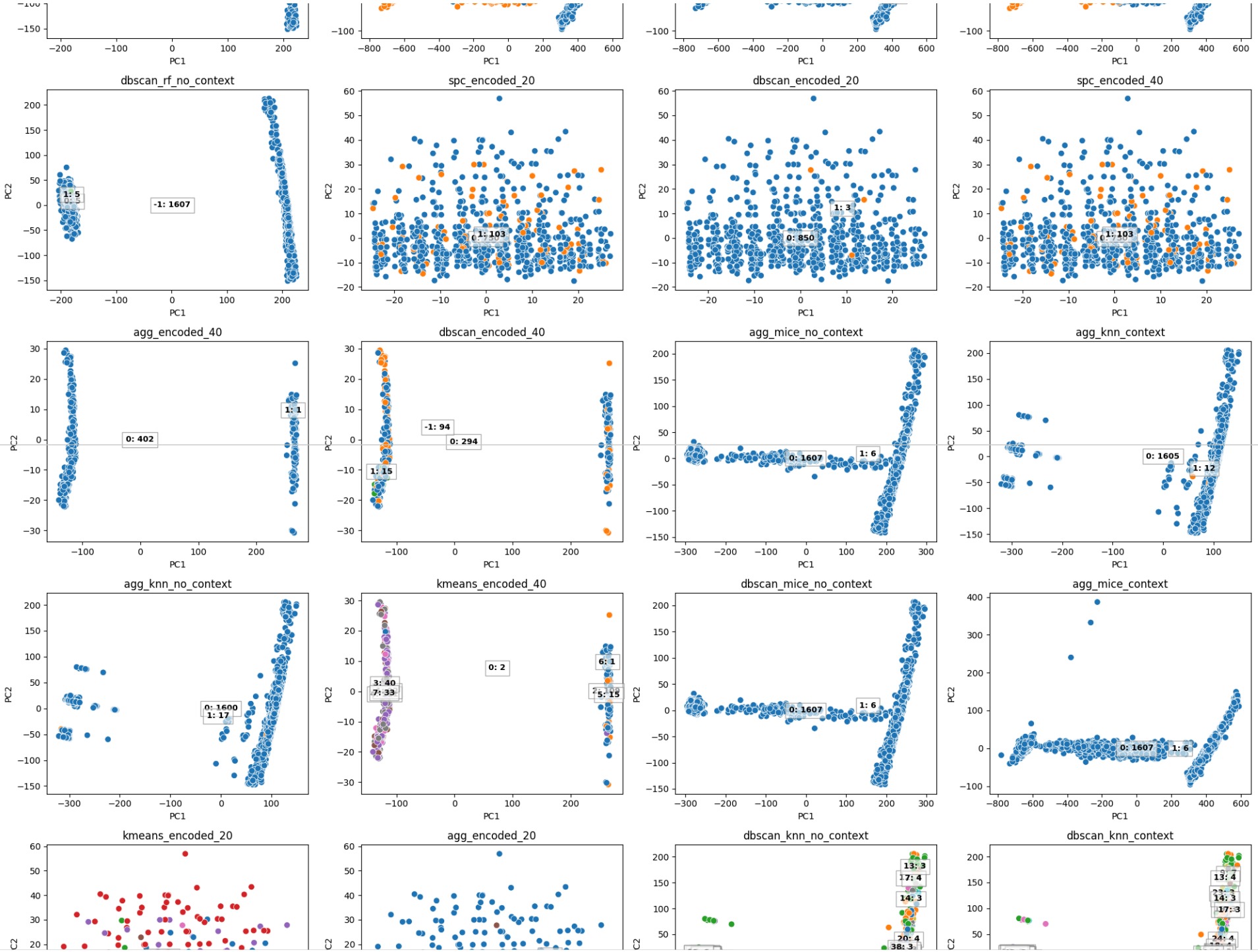
```
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA
import seaborn as sns
import numpy as np

def visualize_clusters(dfs_list, cols=4, random_state=42):
    """
    Visualize multiple clustered DataFrames in a grid of PCA-reduced 2D scatter plots.

    dfs_list: list of tuples like (df, df_name)
    """
    rows = (len(dfs_list) + cols - 1) // cols
    fig, axes = plt.subplots(rows, cols, figsize=(5*cols, 4*rows))
    axes = axes.flatten()

    for ax, (df, df_name) in zip(axes, dfs_list):
        # Infer cluster column: first part of df_name + "_cluster"
        cluster_col = df_name.split("_")[0] + "_cluster"

        if cluster_col not in df.columns:
            print(f"⚠ Cluster column '{cluster_col}' not found in {df_name}, skipping...")
            continue

        # Reduce features to 2D for scatter plot
        X = df.drop(columns=[cluster_col]).values
        pca = PCA(n_components=2, random_state=random_state)
        X_2d = pca.fit_transform(X)

        # Compute cluster counts
        cluster_counts = df[cluster_col].value_counts()

        # Scatter plot with cluster colors
        sns.scatterplot(
            x=X_2d[:,0], y=X_2d[:,1],
            hue=df[cluster_col].astype(str),
            palette="tab10",
            ax=ax,
            legend=False,
            s=50
        )

        # Annotate cluster counts at centroids
        for cluster_id, count in cluster_counts.items():
            mask = df[cluster_col] == cluster_id
            centroid_x = X_2d[mask, 0].mean()
            centroid_y = X_2d[mask, 1].mean()
            ax.text(
                centroid_x, centroid_y,
                f"{cluster_id}: {count}",
```

```
                    fontsize=9, fontweight='bold',
                    ha='center', va='center',
                    bbox=dict(facecolor='white', alpha=0.6, edgecolor='gray')
                )

            ax.set_title(df_name)
            ax.set_xlabel("PC1")
            ax.set_ylabel("PC2")

        # Remove unused axes
        for ax in axes[len(dfs_list):]:
            fig.delaxes(ax)

        plt.tight_layout()
        plt.show()
```

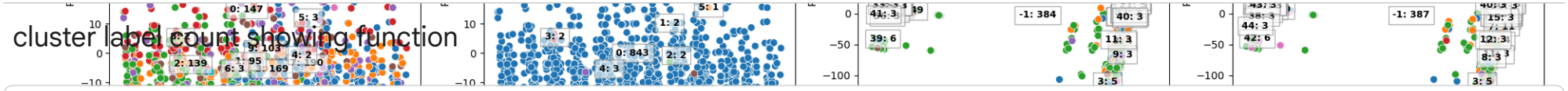
showing all 32 dataframe clusters

```
visualize_clusters(all_dfs_sorted, cols=4)
```


agg_rf_no_context
1: 858
0: 759
kmeans_mice_no_context
1: 731
0: 882
agg_rf_context
0: 858
1: 759
spc_rf_no_context
0: 858
1: 759
dbscan_rf_context
1: 858
0: 759
spc_mice_no_context
1: 730
0: 883
spc_knn_no_context
0: 422
1: 1195
spc_rf_context
0: 858
1: 759
spc_knn_context
0: 422
1: 1195
kmeans_rf_no_context
1: 859
0: 758
kmeans_knn_no_context
1: 432
0: 1185
kmeans_knn_context
1: 432
0: 1185
kmeans_rf_context
1: 859
0: 758
spc_mice_context
1: 818
0: 795
dbscan_mice_context
1: 3
-1: 1607
0: 3
kmeans_mice_context
0: 718
1: 895
PC1
PC2

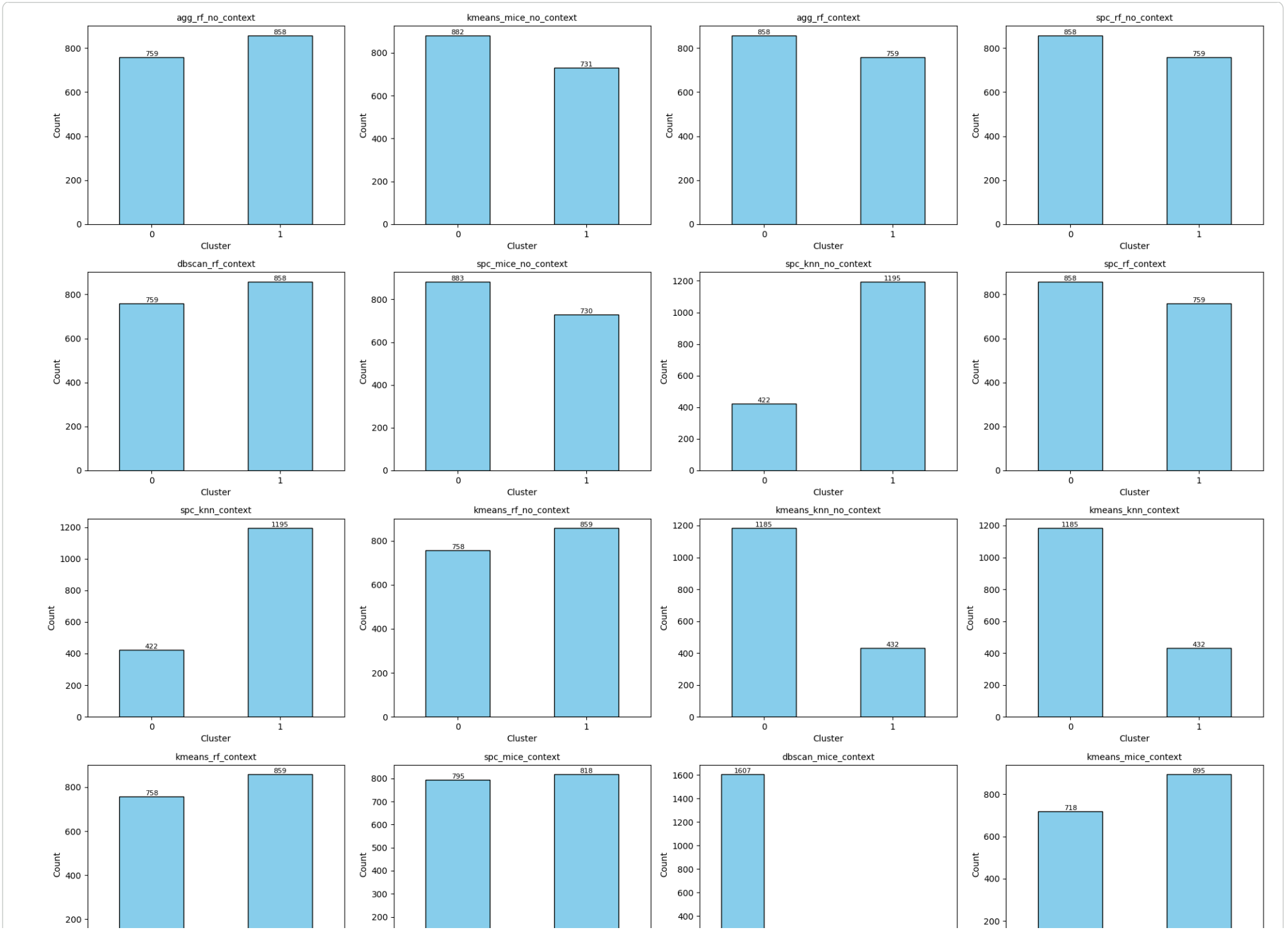

dbscan_rf_no_context
1: 5
0: 5
-1: 1607
spc_encoded_20
1: 103
dbscan_encoded_20
1: 3
0: 850
spc_encoded_40
1: 103
agg_encoded_40
0: 402
1: 1
dbscan_encoded_40
-1: 94
0: 294
1: 15
agg_mice_no_context
0: 1607
1: 6
agg_knn_context
0: 1605
1: 12
agg_knn_no_context
0: 1600
1: 17
kmeans_encoded_40
0: 2
3: 40
7: 33
6: 1
5: 15
dbscan_mice_no_context
0: 1607
1: 6
agg_mice_context
0: 1607
1: 6
kmeans_encoded_20
agg_encoded_20
dbscan_knn_no_context
13: 3
17: 4
14: 3
20: 4
38: 3
dbscan_knn_context
13: 4
14: 3
17: 3
PC1
PC2

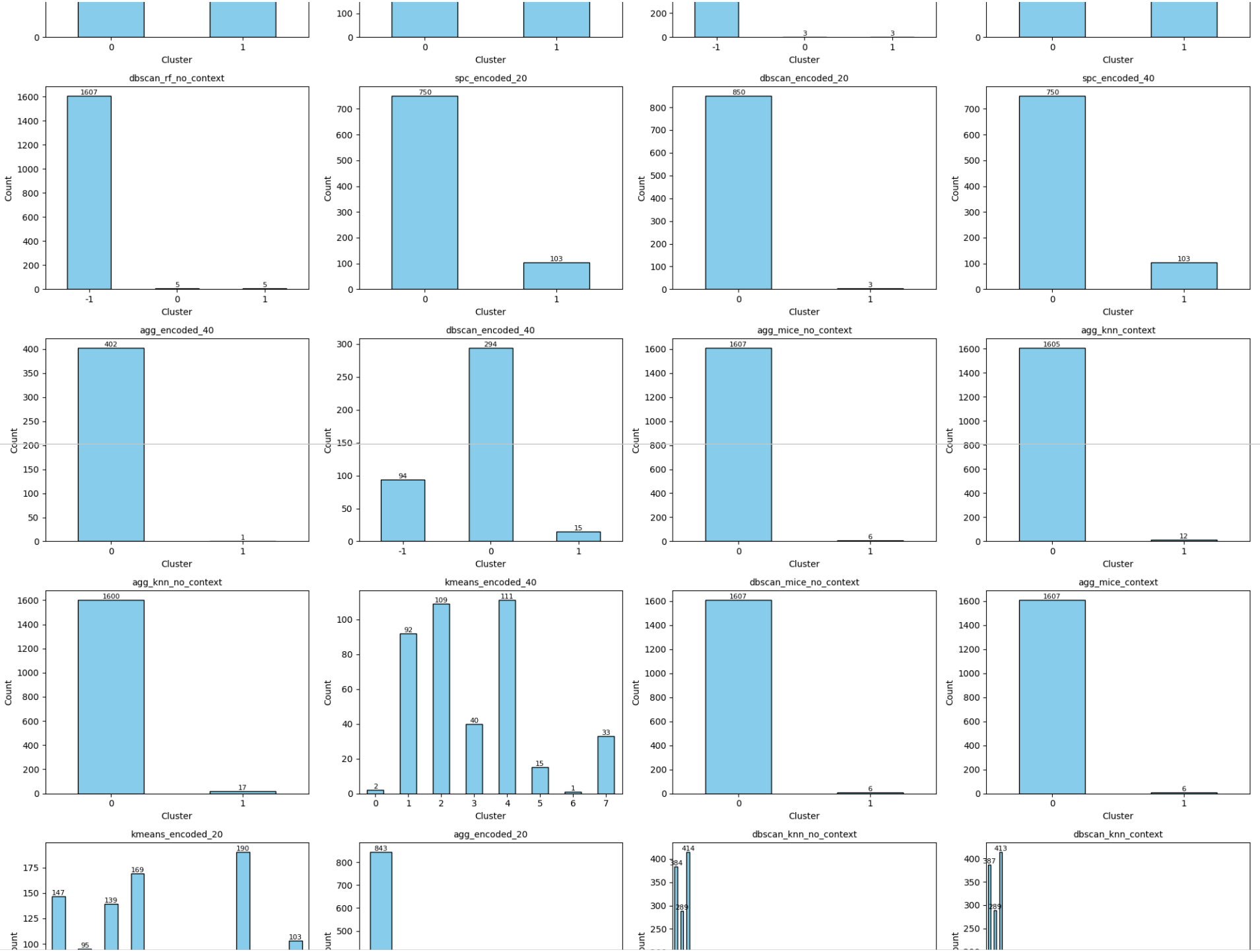
cluster label count showing function

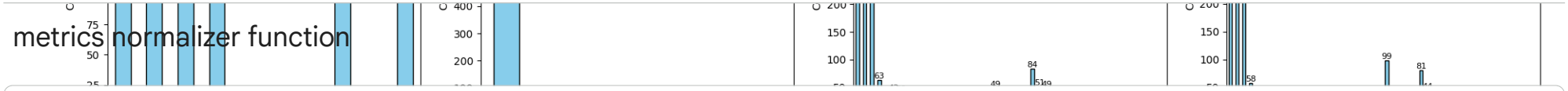



```
import matplotlib.pyplot as plt

def plot_cluster_counts(valid_dfs_list, cols=4):
    all_counts = []

    for df, df_name in valid_dfs_list:
        # Infer cluster column name
        cluster_col = df_name.split("_")[0] + "_cluster"

        if cluster_col in df.columns:
            counts = df[cluster_col].value_counts().sort_index()
            all_counts.append((df_name, counts))
        else:
            print(f"⚠ Warning: Cluster column '{cluster_col}' not found in {df_name}")

    # Setup grid
    rows = (len(all_counts) + cols - 1) // cols
    fig, axes = plt.subplots(rows, cols, figsize=(5*cols, 4*rows))
    axes = axes.flatten()

    # Plot each cluster count
    for ax, (df_name, counts) in zip(axes, all_counts):
        counts.plot(kind="bar", ax=ax, color="skyblue", edgecolor="black")
        ax.set_title(df_name, fontsize=10)
        ax.set_xlabel("Cluster")
        ax.set_ylabel("Count")
        ax.tick_params(axis='x', rotation=0)

        # Annotate counts on top of bars
        for p in ax.patches:
            ax.annotate(
                str(int(p.get_height())),
                (p.get_x() + p.get_width() / 2, p.get_height()),
                ha="center", va="bottom", fontsize=8
            )

    # Remove unused axes
    for ax in axes[len(all_counts):]:
        fig.delaxes(ax)

    plt.tight_layout()
    plt.show()
```

```
plot_cluster_counts(all_dfs_sorted, cols=4)
```


agg_rf_no_context
759
858
kmeans_mice_no_context
882
731
agg_rf_context
858
759
spc_rf_no_context
858
759
dbscan_rf_context
759
858
spc_mice_no_context
883
730
spc_knn_no_context
422
1195
spc_rf_context
858
759
spc_knn_context
422
1195
kmeans_rf_no_context
758
859
kmeans_knn_no_context
1185
432
kmeans_knn_context
1185
432
kmeans_rf_context
758
859
spc_mice_context
795
818
dbscan_mice_context
1607
kmeans_mice_context
718
895
Count
Cluster


dbscan_rf_no_context
Cluster
Count
spc_encoded_20
dbscan_encoded_20
spc_encoded_40
agg_encoded_40
dbscan_encoded_40
agg_mice_no_context
agg_knn_context
agg_knn_no_context
kmeans_encoded_40
dbscan_mice_no_context
agg_mice_context
kmeans_encoded_20
agg_encoded_20
dbscan_knn_no_context
dbscan_knn_context



metrics normalizer function

```
import numpy as np

def normalize_per_metric(per_metric_sorted):
    """
    Normalize all metric values in a per-metric dictionary using min-max scaling.

    Args:
        per_metric_sorted (dict): Dictionary like your example,
            {metric_name: [(dataset_name, value), ...]}

    Returns:
        dict: Normalized dictionary with same structure but values scaled [0, 1].
    """
    normalized_dict = {}

    for metric, values in per_metric_sorted.items():
        # Extract just the metric values
        vals = np.array([v for _, v in values], dtype=float)

        # Handle NaN values
        is_nan = np.isnan(vals)
        finite_vals = vals[~is_nan]

        if finite_vals.size == 0:
            # If all are NaN, keep as NaN
            normalized_vals = vals
        else:
            min_val = finite_vals.min()
            max_val = finite_vals.max()
            # Avoid division by zero if all values are equal
            if max_val == min_val:
                normalized_vals = np.ones_like(vals)
            else:
                normalized_vals = (vals - min_val) / (max_val - min_val)

        # Recreate the list of tuples
        normalized_dict[metric] = [(name, float(val)) for (name, _), val in zip(values, normalized_vals)]

    return normalized_dict
```

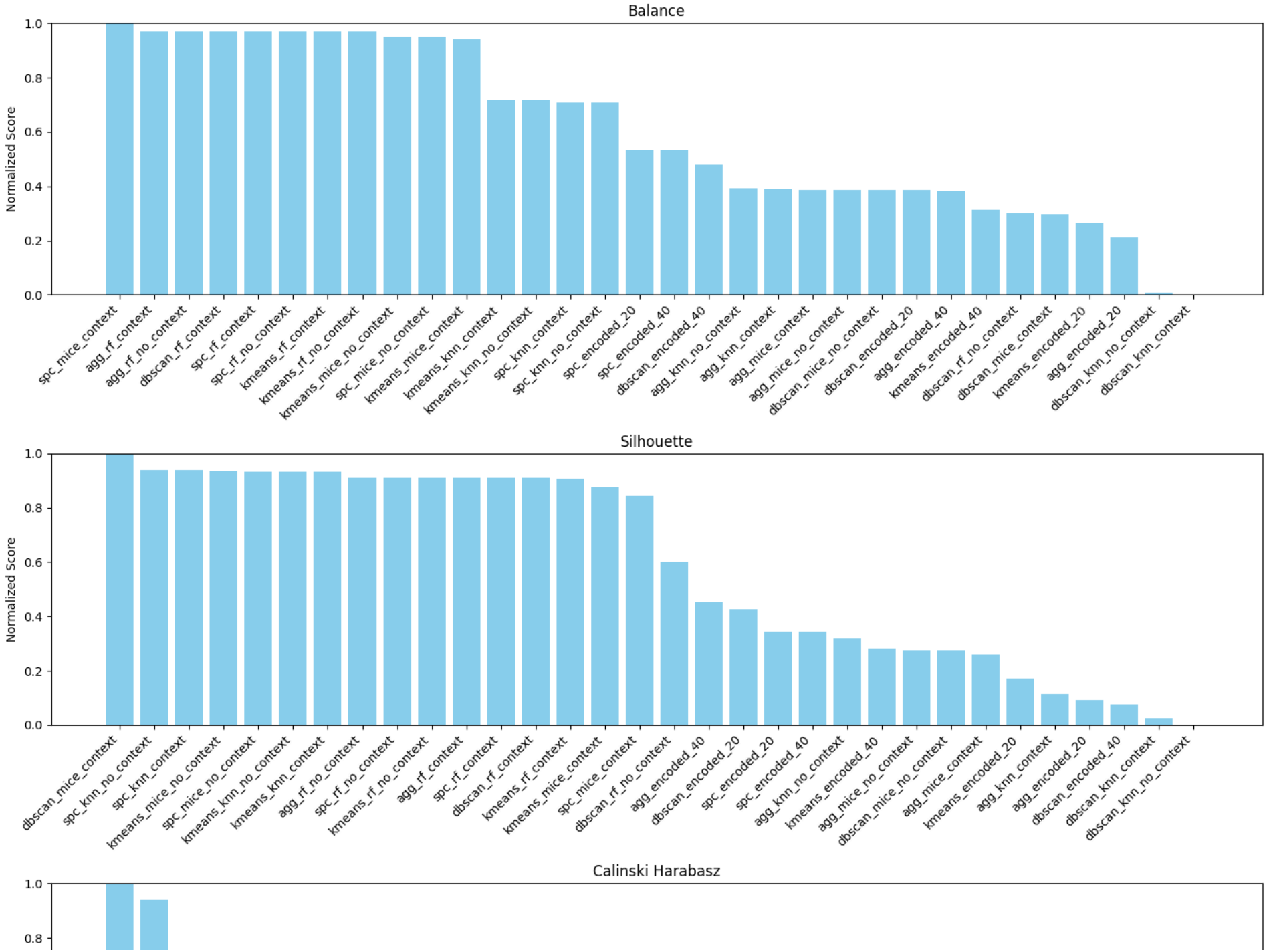
```
normalized_metrics = normalize_per_metric(per_metric_sorted)
```

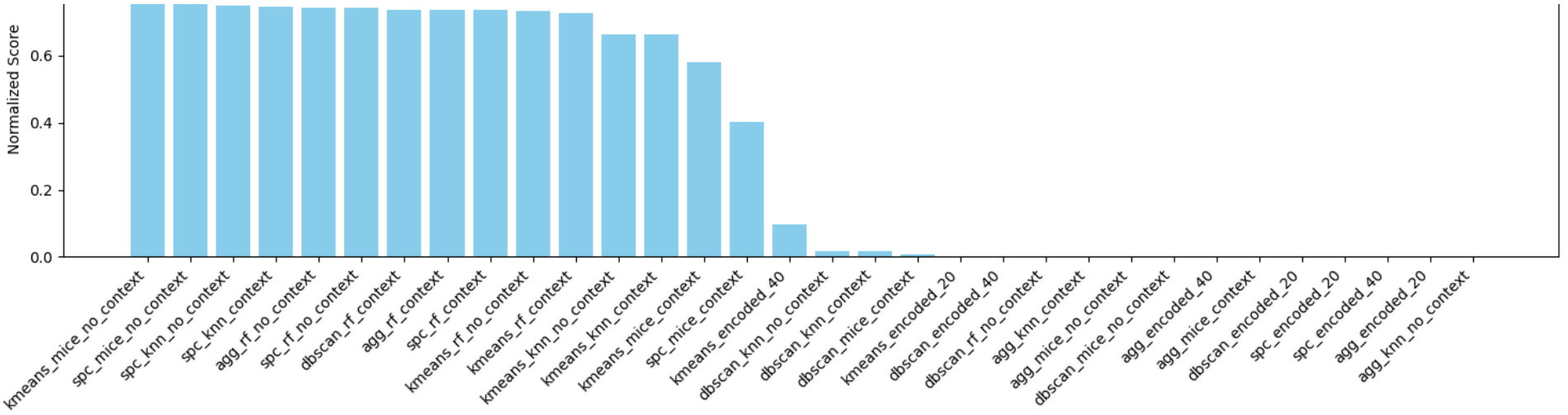
metric comparsion between dfs

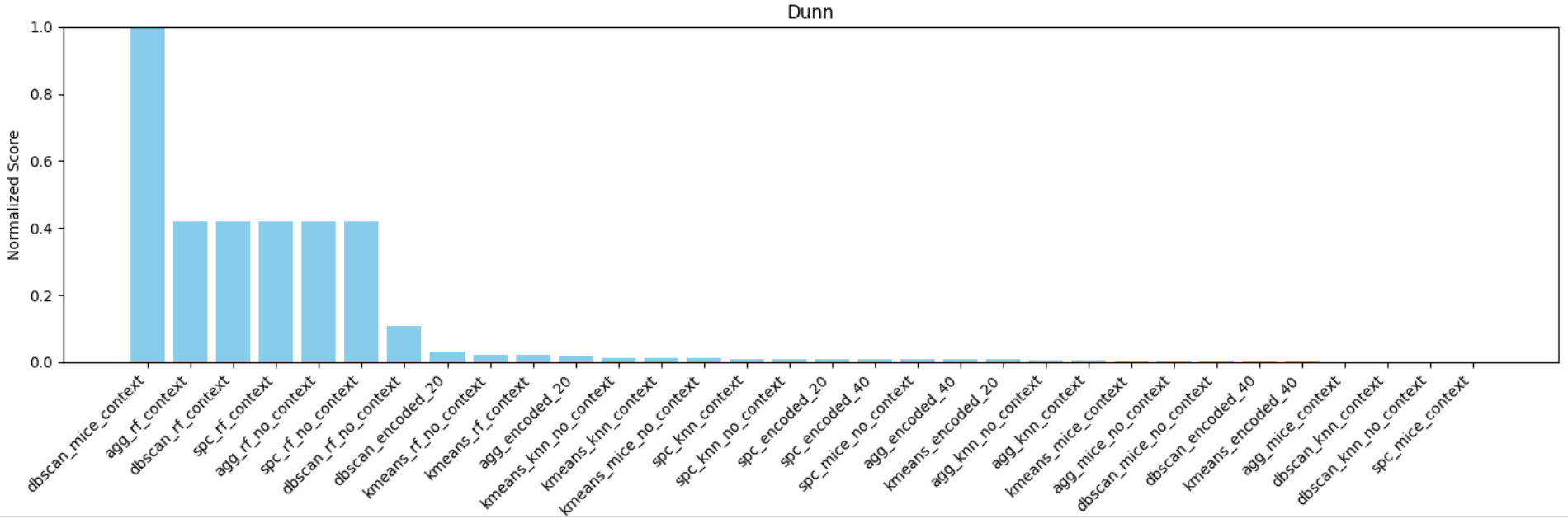
```
import matplotlib.pyplot as plt
import math

def plot_normalized_metrics(normalized_metrics):
```

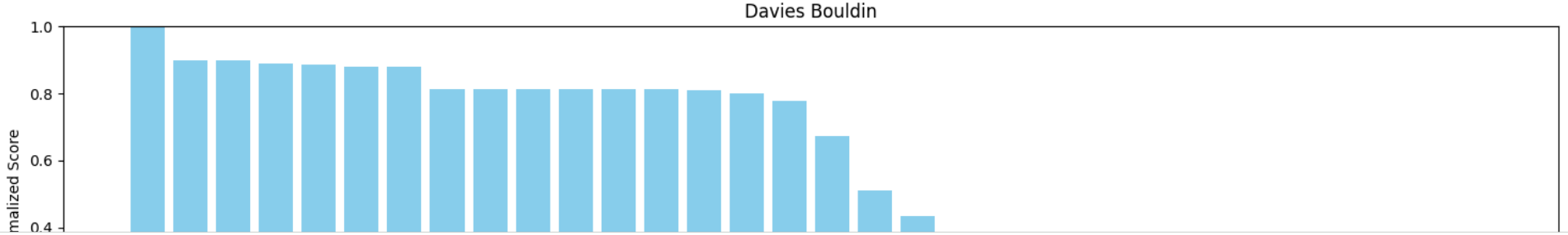
```
    """
    Plot vertical bar charts for normalized metric values for all datasets,
    arranged in 2 columns.

    Args:
        normalized_metrics (dict): {metric_name: [(dataset_name, normalized_value), ...]}
    """
    metrics = list(normalized_metrics.keys())
    n_metrics = len(metrics)

    # Layout: 2 columns
    n_cols = 1
    n_rows = math.ceil(n_metrics / n_cols)

    plt.figure(figsize=(15*n_cols, 5*n_rows))

    for idx, metric in enumerate(metrics, start=1):
        plt.subplot(n_rows, n_cols, idx)

        data = normalized_metrics[metric]
        names = [name for name, _ in data]
        values = [val for _, val in data]

        # Vertical bar plot
        plt.bar(names, values, color='skyblue')
        plt.ylabel("Normalized Score")
        plt.title(metric.replace('_', ' ').title())
        plt.ylim(0, 1)  # normalized values between 0 and 1
        plt.xticks(rotation=45, ha='right')

    plt.tight_layout()
    plt.show()
```

```
plot_normalized_metrics(normalized_metrics)
```


Balance
Normalized Score
1.0
0.8
0.6
0.4
0.2
0.0
spc_mice_context
agg_rf_context
agg_rf_no_context
dbscan_rf_context
spc_rf_context
spc_rf_no_context
kmeans_rf_context
kmeans_rf_no_context
kmeans_mice_no_context
spc_mice_no_context
kmeans_mice_context
kmeans_knn_context
kmeans_knn_no_context
spc_knn_context
spc_knn_no_context
spc_encoded_20
spc_encoded_40
dbscan_encoded_40
agg_knn_no_context
agg_knn_context
agg_mice_context
agg_mice_no_context
dbscan_mice_no_context
dbscan_encoded_20
agg_encoded_40
kmeans_encoded_40
dbscan_rf_no_context
dbscan_mice_context
kmeans_encoded_20
agg_encoded_20
dbscan_knn_no_context
dbscan_knn_context
Silhouette
Normalized Score
1.0
0.8
0.6
0.4
0.2
0.0
dbscan_mice_context
spc_knn_no_context
spc_knn_context
kmeans_mice_no_context
spc_mice_no_context
kmeans_knn_no_context
kmeans_knn_context
agg_rf_no_context
spc_rf_no_context
kmeans_rf_no_context
agg_rf_context
spc_rf_context
dbscan_rf_context
kmeans_rf_context
kmeans_mice_context
spc_mice_context
dbscan_rf_no_context
agg_encoded_40
dbscan_encoded_20
spc_encoded_20
spc_encoded_40
agg_knn_no_context
kmeans_encoded_40
agg_mice_no_context
dbscan_mice_no_context
agg_mice_context
kmeans_encoded_20
agg_knn_context
agg_encoded_20
dbscan_encoded_40
dbscan_knn_context
dbscan_knn_no_context
Calinski Harabasz
1.0
0.8


Normalized Score
0.0
0.2
0.4
0.6
kmeans_mice_no_context
spc_mice_no_context
spc_knn_no_context
spc_knn_context
agg_rf_no_context
spc_rf_no_context
dbscan_rf_context
agg_rf_context
spc_rf_context
kmeans_rf_no_context
kmeans_rf_context
kmeans_knn_no_context
kmeans_knn_context
kmeans_mice_context
spc_mice_context
kmeans_encoded_40
dbscan_knn_no_context
dbscan_knn_context
dbscan_mice_context
kmeans_encoded_20
dbscan_encoded_40
dbscan_rf_no_context
agg_knn_context
agg_mice_no_context
dbscan_mice_no_context
agg_encoded_40
agg_mice_context
dbscan_encoded_20
spc_encoded_20
spc_encoded_40
agg_encoded_20
agg_knn_no_context


Dunn
Normalized Score
0.0
0.2
0.4
0.6
0.8
1.0
dbscan_mice_context
agg_rf_context
dbscan_rf_context
spc_rf_context
agg_rf_no_context
spc_rf_no_context
dbscan_rf_no_context
dbscan_encoded_20
kmeans_rf_no_context
kmeans_rf_context
agg_encoded_20
kmeans_knn_no_context
kmeans_knn_context
kmeans_mice_no_context
spc_knn_context
spc_knn_no_context
spc_encoded_20
spc_encoded_40
spc_mice_no_context
agg_encoded_40
kmeans_encoded_20
agg_knn_no_context
agg_knn_context
kmeans_mice_context
agg_mice_no_context
dbscan_mice_no_context
dbscan_encoded_40
kmeans_encoded_40
agg_mice_context
dbscan_knn_context
dbscan_knn_no_context
spc_mice_context

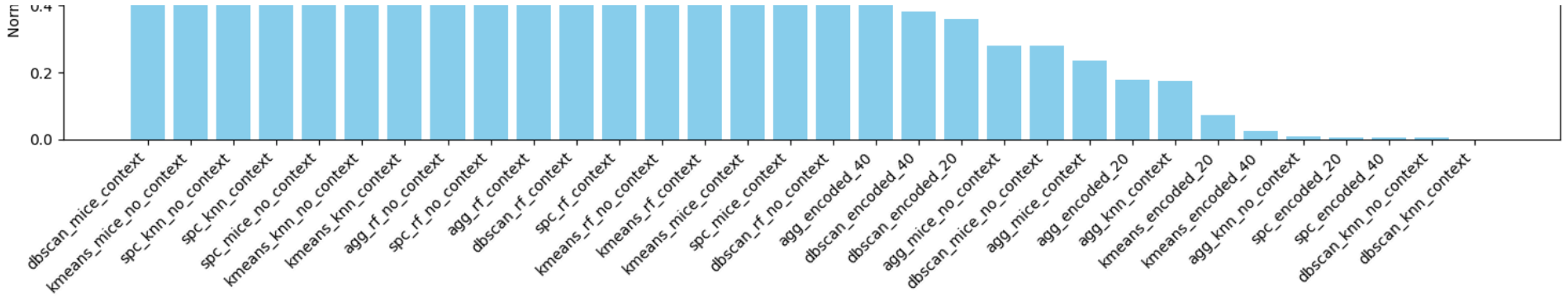

Davies Bouldin
malized Score
0.4
0.6
0.8
1.0

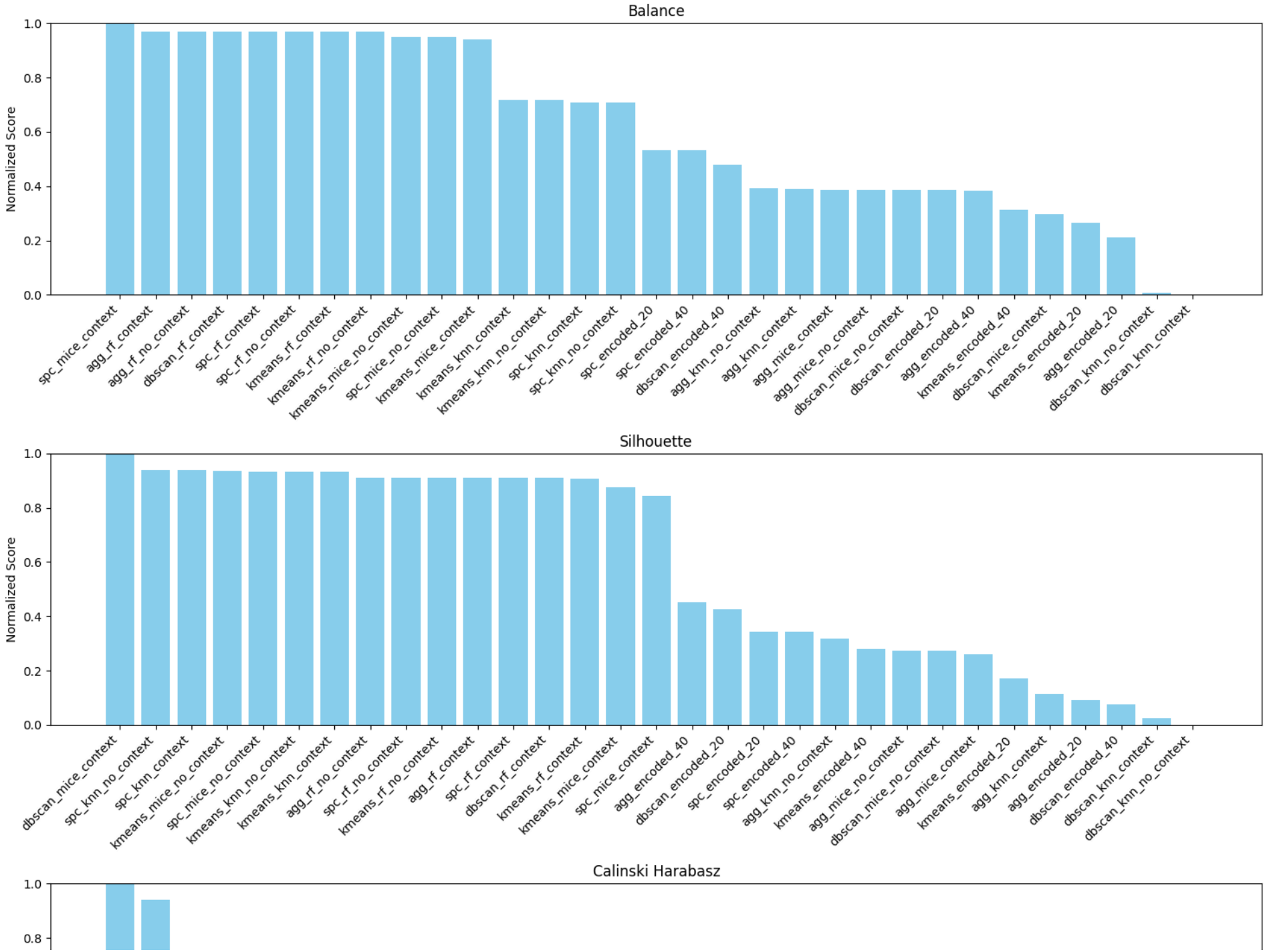
```
def remove_dataset_from_per_metric(per_metric_sorted, dataset_name="dbscan_rf_no_context"):
    """
    Remove a specific dataset from per_metric_sorted dictionary.

    Args:
        per_metric_sorted (dict): {metric_name: [(dataset_name, value), ...]}
        dataset_name (str): Name of the dataset to remove

    Returns:
        dict: New dictionary with the dataset removed
    """
    new_dict = {}
    for metric, data in per_metric_sorted.items():
        new_data = [(name, val) for name, val in data if name != dataset_name]
        new_dict[metric] = new_data
    return new_dict
```

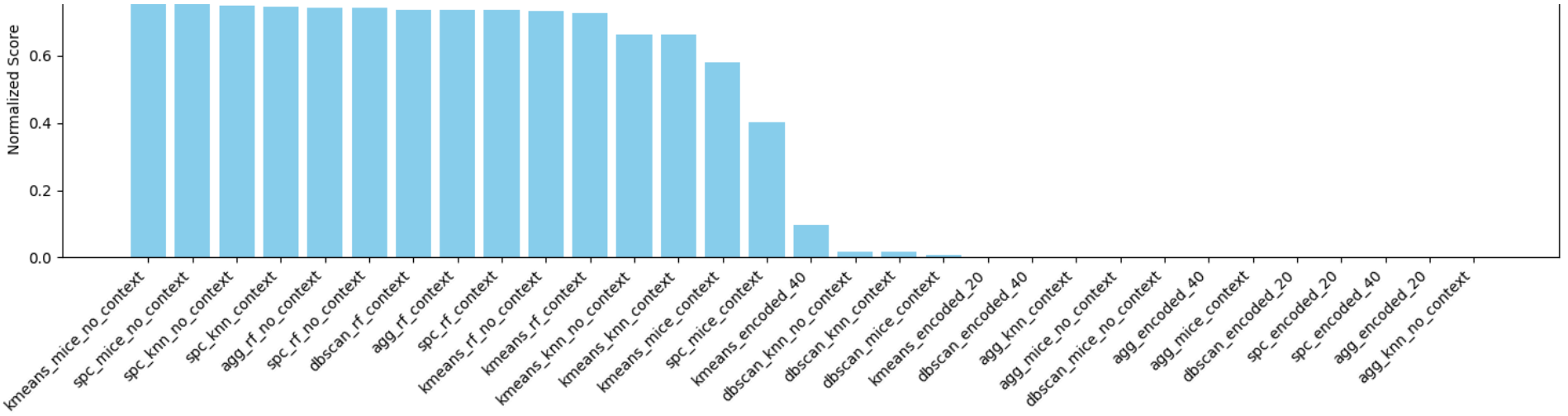
```
per_metric_sorted_clean = remove_dataset_from_per_metric(per_metric_sorted)
```

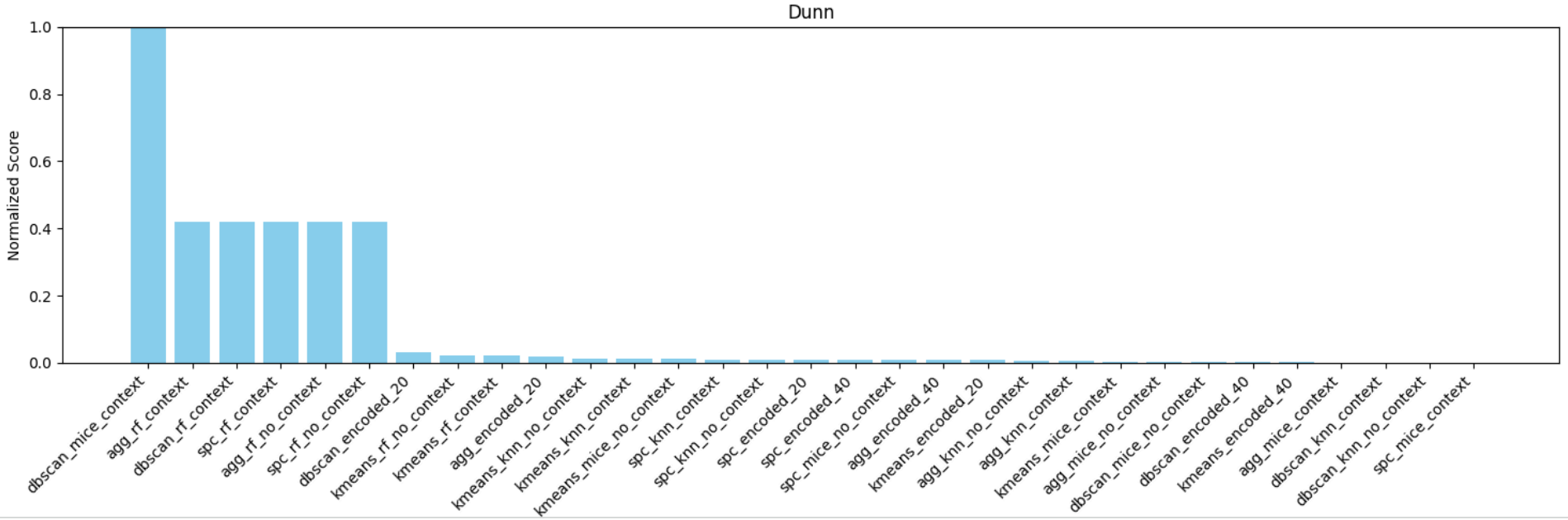
```
clean_normalized_metrics = normalize_per_metric(per_metric_sorted_clean)
```

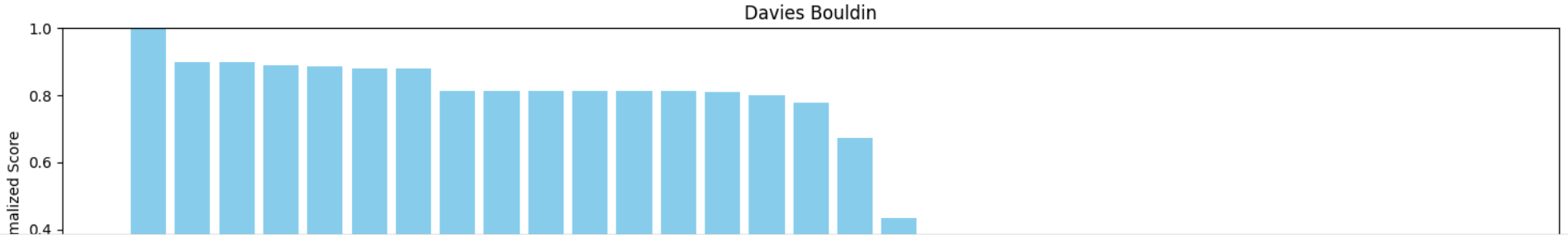
```
plot_normalized_metrics(clean_normalized_metrics)
```




Balance
Normalized Score
1.0
0.8
0.6
0.4
0.2
0.0
spc_mice_context
agg_rf_context
agg_rf_no_context
dbscan_rf_context
spc_rf_context
spc_rf_no_context
kmeans_rf_context
kmeans_rf_no_context
kmeans_mice_no_context
spc_mice_no_context
kmeans_mice_context
kmeans_knn_context
kmeans_knn_no_context
spc_knn_context
spc_knn_no_context
spc_encoded_20
spc_encoded_40
dbscan_encoded_40
agg_knn_no_context
agg_knn_context
agg_mice_context
agg_mice_no_context
dbscan_mice_no_context
dbscan_encoded_20
agg_encoded_40
kmeans_encoded_40
dbscan_mice_context
kmeans_encoded_20
agg_encoded_20
dbscan_knn_no_context
dbscan_knn_context
Silhouette
Normalized Score
1.0
0.8
0.6
0.4
0.2
0.0
dbscan_mice_context
spc_knn_no_context
spc_knn_context
kmeans_mice_no_context
spc_mice_no_context
kmeans_knn_no_context
kmeans_knn_context
agg_rf_no_context
spc_rf_no_context
kmeans_rf_no_context
agg_rf_context
spc_rf_context
dbscan_rf_context
kmeans_rf_context
kmeans_mice_context
spc_mice_context
agg_encoded_40
dbscan_encoded_20
spc_encoded_20
spc_encoded_40
agg_knn_no_context
kmeans_encoded_40
agg_mice_no_context
dbscan_mice_no_context
agg_mice_context
kmeans_encoded_20
agg_knn_context
agg_encoded_20
agg_encoded_40
dbscan_encoded_40
dbscan_knn_context
dbscan_knn_no_context
Calinski Harabasz
1.0
0.8


Normalized Score
0.0
0.2
0.4
0.6
kmeans_mice_no_context
spc_mice_no_context
spc_knn_no_context
spc_knn_context
agg_rf_no_context
spc_rf_no_context
dbscan_rf_context
agg_rf_context
spc_rf_context
kmeans_rf_no_context
kmeans_rf_context
kmeans_knn_no_context
kmeans_knn_context
kmeans_mice_context
spc_mice_context
kmeans_encoded_40
dbscan_knn_no_context
dbscan_knn_context
dbscan_mice_context
kmeans_encoded_20
dbscan_encoded_40
agg_knn_context
agg_mice_no_context
dbscan_mice_no_context
agg_encoded_40
agg_mice_context
dbscan_encoded_20
spc_encoded_20
spc_encoded_40
agg_encoded_20
agg_knn_no_context


Dunn
Normalized Score
0.0
0.2
0.4
0.6
0.8
1.0
dbscan_mice_context
agg_rf_context
dbscan_rf_context
spc_rf_context
agg_rf_no_context
spc_rf_no_context
dbscan_encoded_20
kmeans_rf_no_context
kmeans_rf_context
agg_encoded_20
kmeans_knn_no_context
kmeans_knn_context
kmeans_mice_no_context
spc_knn_context
spc_knn_no_context
spc_encoded_20
spc_encoded_40
spc_mice_no_context
agg_encoded_40
kmeans_encoded_20
agg_knn_no_context
agg_knn_context
kmeans_mice_context
agg_mice_no_context
dbscan_mice_no_context
dbscan_encoded_40
dbscan_encoded_20
kmeans_encoded_40
agg_mice_context
dbscan_knn_context
dbscan_knn_no_context
spc_mice_context


Davies Bouldin
malized Score
0.4
0.6
0.8
1.0

Extra comparison: computing average metrics for groups to compare

```
import numpy as np

def compute_group_average_metrics(all_evaluation_results, df_groups):
    """
    Compute average metrics for each group of datasets.

    Args:
        all_evaluation_results (dict): {dataset_name: {metric_name: value, ...}}
        df_groups (dict): {group_name: list of (df, dataset_name)}

    Returns:
        dict: {group_name: {metric_name: average_value}}
    """
    group_avg_metrics = {}

    for group_name, df_list in df_groups.items():
        metric_sums = {}
        metric_counts = {}

        # Loop through each dataset in the group
        for df, dataset_name in df_list:
            metrics = all_evaluation_results.get(dataset_name)
            if metrics is None:
                continue

            for metric, value in metrics.items():
                if np.isnan(value):
                    continue
                metric_sums[metric] = metric_sums.get(metric, 0) + value
                metric_counts[metric] = metric_counts.get(metric, 0) + 1

        # Compute average
        group_avg_metrics[group_name] = {}
        for metric in metric_sums:
            group_avg_metrics[group_name][metric] = metric_sums[metric] / metric_counts[metric]

    return group_avg_metrics
```

```
# Define your groups as a dictionary
df_groups = {
    "agg": agg_dfs,
    "dbscan": dbscan_dfs,
    "kmeans": kmeans_dfs,
    "spc": spc_dfs,
    "no_context": no_context_dfs,
    "context": context_dfs,
    "no_imputed": no_imputed_dfs,
    "mice": mice_dfs,
    "rf": rf_dfs,
    "knn": knn_dfs,
```

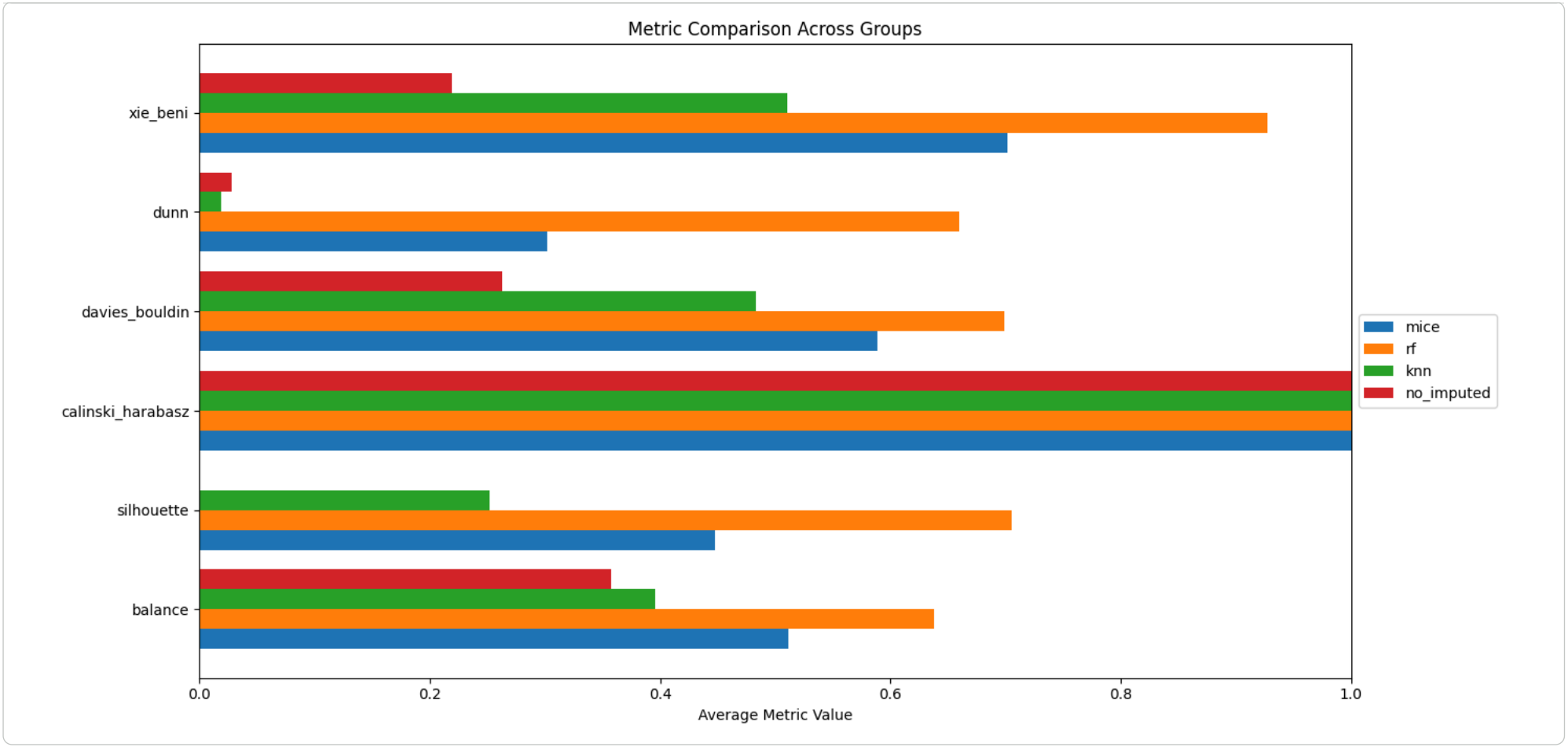
```
}

group_avg_metrics = compute_group_average_metrics(all_evaluation_results, df_groups)
```

showing groups comparison bar chart

```
import matplotlib.pyplot as plt
import numpy as np

def compare_group_metrics(group_avg_metrics, selected_groups, width=14, height=5):
    """
    Compare metrics between multiple groups using horizontal grouped bar chart.

    Args:
        group_avg_metrics (dict): {group_name: {metric_name: average_value}}
        selected_groups (list): List of group names to compare (must be keys in group_avg_metrics)
    """
    metrics = list(next(iter(group_avg_metrics.values())).keys())  # list of metrics
    n_metrics = len(metrics)
    n_groups = len(selected_groups)

    # Bar positions
    y = np.arange(n_metrics)
    bar_height = 0.8 / n_groups  # split each metric bar

    plt.figure(figsize=(width, height))

    # Plot each group's bars
    for i, group in enumerate(selected_groups):
        values = [group_avg_metrics[group][metric] for metric in metrics]
        plt.barh(y + i*bar_height, values, height=bar_height, label=group)

    plt.yticks(y + bar_height*(n_groups-1)/2, metrics)  # center metric labels
    plt.xlabel("Average Metric Value")
    plt.title("Metric Comparison Across Groups")
    plt.legend(loc='center left', bbox_to_anchor=(1, 0.5))  # legend to the right
    plt.xlim(0, 1)  # assuming normalized metrics; adjust if not
    plt.tight_layout()
    plt.show()
```

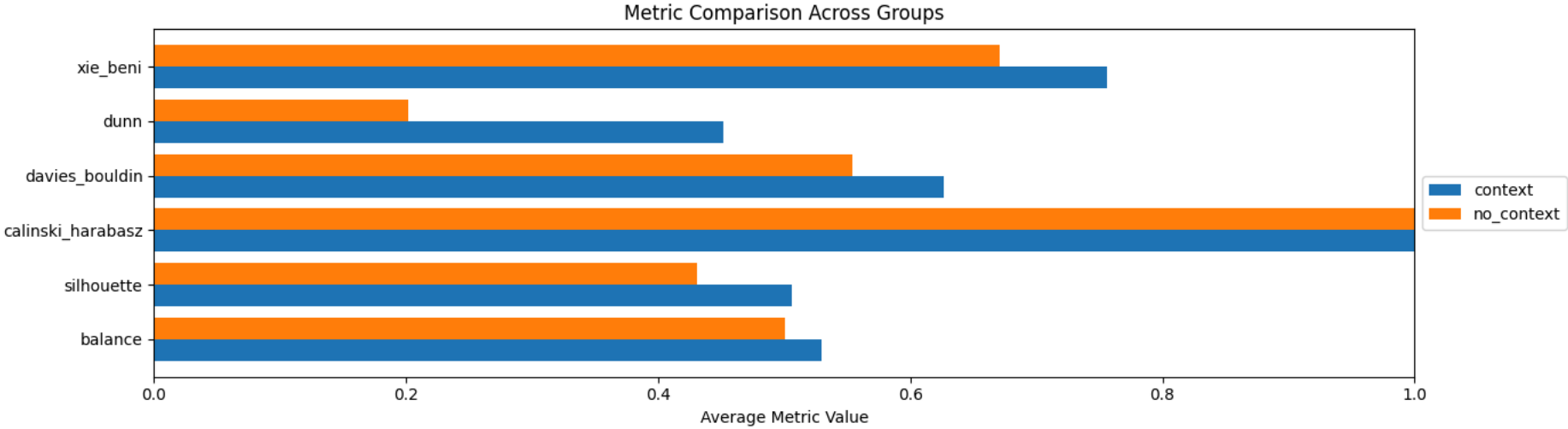
comparison between imputation

```
compare_group_metrics(group_avg_metrics, ["mice", "rf", "knn", "no_imputed"], height=7)
```



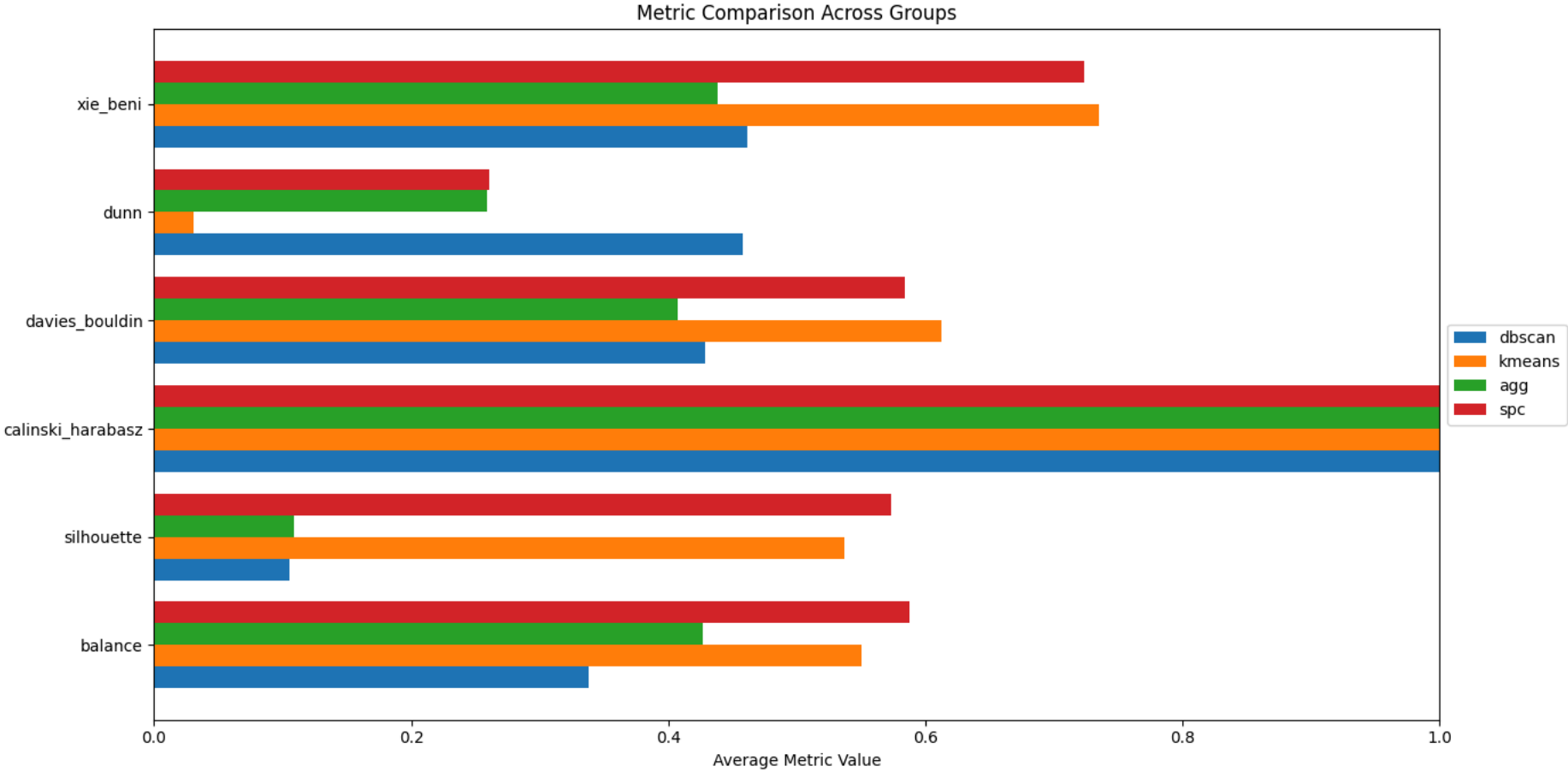
comparing contextual or non contextual imputaion

```
compare_group_metrics(group_avg_metrics, ["context", "no_context"], height=4)
```



comparing clustering alorthms

```
compare_group_metrics(group_avg_metrics, ["dbscan", "kmeans", "agg", "spc"], height=7)
```




```
import pandas as pd

def create_metrics_table(per_metric_sorted, score_map):
    """
    Creates a pandas DataFrame from the per_metric_sorted dictionary and sorts it by composite score.

    Args:
        per_metric_sorted (dict): {metric_name: [(dataset_name, value), ...]}
        score_map (dict): {dataset_name: composite_score}

    Returns:
        pd.DataFrame: A DataFrame with dataset names as index and metrics as columns, sorted by composite score.
    """
    # Get all unique dataset names
    dataset_names = sorted(list(set([name for metric_data in per_metric_sorted.values() for name, _ in metric_data])))

    # Create a dictionary to store metrics for each dataset
    data = {name: {} for name in dataset_names}
```

```
        # Populate the dictionary
        for metric, metric_data in per_metric_sorted.items():
            for name, value in metric_data:
                data[name][metric] = value

        # Create the DataFrame
        df_metrics = pd.DataFrame.from_dict(data, orient='index')

        # Sort columns based on METRIC_ORDER
        sorted_columns = [m for m in METRIC_ORDER if m in df_metrics.columns]
        df_metrics = df_metrics[sorted_columns]

        # Add composite score as a column for sorting and display
        df_metrics['composite_score'] = df_metrics.index.map(score_map)

        # Sort by composite score in descending order
        df_metrics = df_metrics.sort_values(by='composite_score', ascending=False)

        # The composite score column will now be kept in the output
        return df_metrics

    # Create and display the table, passing score_map
    df_metrics_table = create_metrics_table(per_metric_sorted, score_map)
    display(df_metrics_table)
```

	balance	silhouette	calinski_harabasz	dunn	davies_bouldin	xie_beni	composite_score
agg_rf_no_context	0.685461	0.756380	9792.497921	0.980954	0.730318	0.960545	27.65
kmeans_mice_no_context	0.674009	0.792085	13154.910479	0.030805	0.791936	0.970546	26.95
agg_rf_context	0.685461	0.754761	9686.611574	0.982102	0.729068	0.960131	26.70
spc_rf_no_context	0.685461	0.756380	9792.497921	0.980954	0.730318	0.960545	25.95
dbscan_rf_context	0.685461	0.754761	9686.611574	0.982102	0.729068	0.960131	25.50
spc_mice_no_context	0.673571	0.787801	12376.439745	0.021246	0.789193	0.968759	25.45
spc_knn_no_context	0.538092	0.795369	9875.866564	0.025660	0.800913	0.969425	24.80
spc_rf_context	0.685461	0.754761	9686.611574	0.982102	0.729068	0.960131	24.80
spc_knn_context	0.538092	0.794904	9833.815232	0.025660	0.800527	0.969298	24.70
kmeans_rf_no_context	0.685023	0.755367	9654.365485	0.055459	0.729813	0.960009	23.65
kmeans_knn_no_context	0.542465	0.785370	8744.775850	0.032461	0.786055	0.965104	22.75
kmeans_knn_context	0.542465	0.784890	8708.895980	0.032461	0.785642	0.964965	22.45
kmeans_rf_context	0.685023	0.753743	9549.182978	0.054177	0.728559	0.959586	22.45
spc_mice_context	0.702065	0.670781	5302.263750	0.001820	0.701281	0.929418	20.85
dbscan_mice_context	0.306703	0.874364	96.547161	2.327724	0.883905	0.989749	18.95
kmeans_mice_context	0.668310	0.708758	7649.178141	0.011035	0.718630	0.950553	18.70
dbscan_rf_no_context	0.307709	0.351424	4.619301	0.256321	0.481121	0.697853	12.45
spc_encoded_20	0.438937	0.012104	1.372098	0.025583	0.061092	0.014959	11.90
dbscan_encoded_20	0.356040	0.121859	1.690365	0.071764	0.284405	0.361745	11.40
spc_encoded_40	0.438937	0.012104	1.372098	0.025583	0.061092	0.014959	10.90
agg_encoded_40	0.355308	0.157153	2.160973	0.020834	0.613707	0.685253	10.80
dbscan_encoded_40	0.409134	-0.339146	8.739920	0.008877	0.418675	0.381333	10.40
agg_mice_no_context	0.356184	-0.077905	2.344871	0.009983	0.281811	0.282000	9.85
agg_knn_context	0.358801	-0.285265	2.609730	0.013441	0.272135	0.179908	9.65
kmeans_encoded_40	0.315207	-0.071859	1281.887066	0.008478	0.168569	0.032546	9.35
agg_knn_no_context	0.360987	-0.019878	0.278632	0.016553	0.068518	0.016314	9.35
dbscan_mice_no_context	0.356184	-0.077905	2.344871	0.009983	0.281811	0.282000	8.85
agg_mice_context	0.356184	-0.094494	1.874077	0.004522	0.259546	0.238908	8.45
kmeans_encoded_20	0.289369	-0.212661	28.041057	0.020631	0.190869	0.079326	7.40
agg_encoded_20	0.258243	-0.316824	1.109202	0.043354	0.303899	0.182562	5.75
dbscan_knn_no_context	0.143811	-0.436810	239.623244	0.003395	0.178531	0.012615	4.75
dbscan_knn_context	0.139237	-0.405025	220.061266	0.003395	0.174882	0.007614	4.45

Next steps: Generate code with df_metrics_table | New interactive sheet

Ending

```
# Find the dataframe with the name 'agg_rf_no_context' in the sorted list
agg_rf_no_context_df = None
for df, name in all_dfs_sorted:
    if name == 'agg_rf_no_context':
        agg_rf_no_context_df = df
        break

# Check if the dataframe was found and display its head
if agg_rf_no_context_df is not None:
    display(agg_rf_no_context_df.head())
else:
    print("DataFrame 'agg_rf_no_context' not found in all_dfs_sorted.")
```

	age	age_group	gender	marital_status	profession_group	workplace	religion	financial_condition	hometown	latitude	...	wind_deg	clouds_sky	weather_main	weather_des
0	17.99	5	1	5	34	0	5	0	18	23.960600	...	320.0	40.0	3	
1	20.00	5	1	4	25	1	5	2	143	24.899220	...	116.0	33.0	1	
2	14.00	5	0	6	26	2	5	2	86	23.832984	...	190.0	75.0	2	
3	35.87	3	0	4	26	2	5	2	17	24.850066	...	186.0	100.0	6	
4	18.00	5	0	5	26	2	5	2	31	23.764386	...	154.0	40.0	3	

5 rows × 30 columns

DEcoding teh encoded dfs

```
non_encoded = pd.read_excel("/content/drive/MyDrive/capstone/datasets/4_non_encoded.xlsx")
rf_no_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/11_forest_no_context.xlsx")

# mice_no_context = pd.read_excel("/content/drive/MyDrive/capstone/datasets/7_mice_no_context.xlsx")
```

```
agg_rf_no_context.head()
```

	age	age_group	gender	marital_status	profession_group	workplace	religion	financial_condition	hometown	latitude	...	wind_deg	clouds_sky	weather_main	weather_des
0	17.99	5	1	5	34	0	5	0	18	23.960600		320.0	40.0	3	

```
import json
import pandas as pd
import numpy as np
from sklearn.preprocessing import LabelEncoder
import os

# 1. Load the saved LabelEncoders JSON
drive_path = '/content/drive/MyDrive/capstone/'
filename = 'label_encoders.json'
full_path = os.path.join(drive_path, filename)

with open(full_path, 'r') as f:
    label_encoders_serializable = json.load(f)

# 2. Decode the encoded DataFrame
rf_no_context_decoded = rf_no_context.copy()

for col, classes in label_encoders_serializable.items():
    if col in rf_no_context_decoded.columns:
        le = LabelEncoder()
        le.classes_ = np.array(classes)  # convert list to numpy array
        rf_no_context_decoded[col] = le.inverse_transform(rf_no_context_decoded[col])

print("Decoding completed!")
```

```
Decoding completed!
```

```
rf_no_context_decoded.head()
```

	age	age_group	gender	marital_status	profession_group	workplace	religion	financial_condition	hometown	latitude	...	wind_speed	wind_deg	clouds_sky	weather_ma
0	17.99	teen	male	relationship	worker	rural	muslim	indebted	brahmanbaria	23.960600	...	3.0	320.0	40.0	ha
1	20.00	teen	male	rejected	stportsman	semi-urban	muslim	middle	sylhet	24.899220	...	3.0	116.0	33.0	clou
2	14.00	teen	female	single	student	urban	muslim	middle	manikganj	23.832984	...	3.6	190.0	75.0	drizz
3	35.87	middle-aged	female	rejected	student	urban	muslim	middle	bogura	24.850066	...	3.0	186.0	100.0	ra
4	18.00	teen	female	relationship	student	urban	muslim	middle	dhaka	23.764386	...	3.0	154.0	40.0	ha

5 rows × 29 columns

```
rf_no_context_merged = rf_no_context_decoded.merge(
    agg_rf_no_context[["agg_cluster"]],
    left_index=True,
    right_index=True,
    how="left"
)
```

```
summary = pd.DataFrame({
    "Column": rf_no_context_merged.columns,
    "Data Type": rf_no_context_merged.dtypes.values,
    "Example Values": [
        rf_no_context_merged[col].dropna().unique()[:3].tolist()
        for col in rf_no_context_merged.columns
    ]
})

summary
```

	Column	Data Type	Example Values
0	age	float64	[17.99, 20.0, 14.0]
1	age_group	object	[teen, middle-aged, adult]
2	gender	object	[male, female, third gender]
3	marital_status	object	[relationship, rejected, single]
4	profession_group	object	[worker, stportsman, student]
5	workplace	object	[rural, semi-urban, urban]
6	religion	object	[muslim, hindu, buddhist]
7	financial_condition	object	[indebted, middle, lower]
8	hometown	object	[brahmanbaria, sylhet, manikganj]
9	latitude	float64	[23.9605999, 24.89922, 23.8329836]
10	longitude	float64	[91.1190893, 91.8685271, 89.9666878]
11	reason	object	[harassment, relationship, marital]
12	time	object	[afternoon, night, morning]
13	method	object	[jumping, hanging, overdose]
14	feels_like	float64	[300.65, 299.17, 298.55]
15	temp_min	float64	[299.15, 295.57, 300.65]
16	temp_max	float64	[299.15, 295.57, 300.65]
17	air_pressure	float64	[1008.0, 1012.0, 1004.0]
18	air_humidity	float64	[83.0, 96.0, 78.0]
19	wind_speed	float64	[3.0, 3.6, 3.05]
20	wind_deg	float64	[320.0, 116.0, 190.0]
21	clouds_sky	float64	[40.0, 33.0, 75.0]
22	weather_main	object	[haze, clouds, drizzle]
23	weather_description	object	[haze, scattered clouds, drizzle]
24	suicide_day	float64	[30.0, 11.0, 25.0]
25	suicide_month	float64	[10.0, 4.0, 9.0]
26	suicide_year	float64	[2020.0, 2022.0, 2021.77]
27	suicide_week	float64	[44.0, 15.0, 39.0]
28	suicide_weekday	object	[Friday, Saturday, Thursday]
29	agg_cluster	int64	[0, 1]

Next steps: Generate code with summary | New interactive sheet

```
import pandas as pd

# ------------------------------
# Step 1: Select your dataset
# ------------------------------
top_df, top_name = rf_no_context_merged, "agg_rf_no_context"
cluster_col = "agg_cluster"

# ------------------------------
# Step 2: Define feature groups
# ------------------------------
num_features = [
    "age", "feels_like", "temp_min", "temp_max",
    "air_pressure", "air_humidity", "wind_speed", "clouds_sky"
]

cat_features = [
    "age_group", "gender", "marital_status", "profession_group",
    "workplace", "religion", "financial_condition", "reason",
    "time", "method", "weather_main", "weather_description",
    "suicide_weekday"
]

# ------------------------------
# Step 3: Numerical summary
# ------------------------------
num_summary = top_df.groupby(cluster_col)[num_features].agg(["mean", "median", "count"])

# ------------------------------
# Step 4: Categorical summary (mode + %)
# ------------------------------
cat_summary = {}
for col in cat_features:
    modes = top_df.groupby(cluster_col)[col].agg(
        lambda x: x.value_counts().index[0] if len(x.value_counts()) > 0 else None
    )
    mode_perc = top_df.groupby(cluster_col)[col].agg(
        lambda x: (x.value_counts().iloc[0] / len(x) * 100) if len(x.value_counts()) > 0 else None
    )
    cat_summary[col] = [f"{modes[c]} ({mode_perc[c]:.1f}%)" for c in modes.index]

cat_summary_df = pd.DataFrame(cat_summary, index=modes.index)

# ------------------------------
# Step 5: Display nicely in Jupyter
# ------------------------------
pd.set_option("display.max_columns", None)
pd.set_option("display.width", 2000)
pd.set_option("display.max_colwidth", None)

print(f"📊 Cluster profiling for: {top_name}")

print("\n🔹 Numerical Feature Summary:")
display(num_summary)

print("\n🔹 Categorical Feature Summary (mode + %):")
```

```
display(cat_summary_df)

# ------------------------------
# Step 6: Save to Excel (for paper)
# ------------------------------
output_file = f"{top_name}_cluster_profiling.xlsx"
with pd.ExcelWriter(output_file) as writer:
    num_summary.to_excel(writer, sheet_name="Numerical_Summary")
    cat_summary_df.to_excel(writer, sheet_name="Categorical_Summary")

print(f"\n✅ Cluster profiling tables saved to '{output_file}'")
```

```
📊 Cluster profiling for: agg_rf_no_context

🔹 Numerical Feature Summary:
```

	age			feels_like			temp_min			temp_max			air_pressure			air_humidity			wind_sp
	mean	median	count	mean	median	count	mean	median	count	mean	median	count	mean	median	count	mean	median	count	mean
agg_cluster																			
0	27.996140	24.0	759	299.893900	300.6500	759	298.946100	299.510	759	298.946100	299.5100	759	1004.773386	1004.000	759	85.187088	87.000	759	3.521950
1	25.795315	23.0	858	290.455384	290.4434	858	23.776781	25.606	858	31.797987	32.5425	858	1007.013161	1005.901	858	71.705163	70.115	858	7.266621

```
🔹 Categorical Feature Summary (mode + %):
```

	age_group	gender	marital_status	profession_group	workplace	religion	financial_condition	reason	time	method	weather_main	weather_description	suicide_
agg_cluster													
0	teen (48.5%)	female (54.8%)	married (48.4%)	student (37.0%)	semi-urban (54.5%)	muslim (87.5%)	lower (54.3%)	family issue (23.1%)	night (36.9%)	hanging (73.3%)	clouds (38.1%)	haze (19.6%)	Saturda
1	teen (39.2%)	male (50.8%)	married (47.6%)	student (39.7%)	rural (50.0%)	muslim (85.4%)	lower (57.6%)	mental (26.5%)	night (35.5%)	hanging (74.1%)	partly cloudy (99.9%)	haze (64.7%)	Sunda

```
✅ Cluster profiling tables saved to 'agg_rf_no_context_cluster_profiling.xlsx'
```

Next steps: Generate code with cat_summary_df | New interactive sheet

```
# Step 2: Compare context vs no-context groups
context_scores = [score_map[name] for _, name in context_dfs if name in score_map]
no_context_scores = [score_map[name] for _, name in no_context_dfs if name in score_map]

print("✅ Average Composite Score (Context):", np.mean(context_scores))
print("✅ Average Composite Score (No-Context):", np.mean(no_context_scores))
```

```
✅ Average Composite Score (Context): 18.97083333333333
✅ Average Composite Score (No-Context): 18.537499999999998
```

```
# Step 3: Compare imputation strategies
def avg_score_for_group(group):
    return np.mean([score_map[name] for _, name in group if name in score_map])
```

```
print("RF Imputation Avg Score:", avg_score_for_group(rf_dfs))
print("KNN Imputation Avg Score:", avg_score_for_group(knn_dfs))
print("MICE Imputation Avg Score:", avg_score_for_group(mice_dfs))
print("No Imputation Avg Score:", avg_score_for_group(no_imputed_dfs))
```

```
RF Imputation Avg Score: 23.64375
KNN Imputation Avg Score: 15.3625
MICE Imputation Avg Score: 17.256249999999998
No Imputation Avg Score: 9.737499999999999
```

```
# Step 4: Examine role of balance metric
balance_values = [(name, all_evaluation_results[name]["balance"], score_map[name])
                  for name in all_evaluation_results.keys()]

balance_df = pd.DataFrame(balance_values, columns=["Dataset", "Balance", "Composite"])
balance_df = balance_df.sort_values("Composite", ascending=False)

print("📊 Top vs bottom models by Balance")
display(balance_df.head(10))
display(balance_df.tail(10))
```

📊 Top vs bottom models by Balance

	Dataset	Balance	Composite
7	agg_rf_no_context	0.685461	27.65
21	kmeans_mice_no_context	0.674009	26.95
6	agg_rf_context	0.685461	26.70
31	spc_rf_no_context	0.685461	25.95
14	dbscan_rf_context	0.685461	25.50
29	spc_mice_no_context	0.673571	25.45
27	spc_knn_no_context	0.538092	24.80
30	spc_rf_context	0.685461	24.80
26	spc_knn_context	0.538092	24.70
23	kmeans_rf_no_context	0.685023	23.65

	Dataset	Balance	Composite
5	agg_mice_no_context	0.356184	9.85
2	agg_knn_context	0.358801	9.65
17	kmeans_encoded_40	0.315207	9.35
3	agg_knn_no_context	0.360987	9.35
13	dbscan_mice_no_context	0.356184	8.85
4	agg_mice_context	0.356184	8.45
16	kmeans_encoded_20	0.289369	7.40
0	agg_encoded_20	0.258243	5.75
11	dbscan_knn_no_context	0.143811	4.75
10	dbscan_knn_context	0.139237	4.45

```
# Step 5: Best model per clustering algorithm
best_per_algo = {}
for algo, group in [("Agglomerative", agg_dfs),
                    ("KMeans", kmeans_dfs),
                    ("DBSCAN", dbscan_dfs),
                    ("Spectral", spc_dfs)]:
    scores = [(name, score_map[name]) for _, name in group if name in score_map]
    best_per_algo[algo] = max(scores, key=lambda x: x[1])

print("🏆 Best model per clustering algorithm:")
for algo, (name, score) in best_per_algo.items():
    print(f"{algo}: {name} → Score {score:.2f}")
```

```
🏆 Best model per clustering algorithm:
Agglomerative: agg_rf_no_context → Score 27.65
KMeans: kmeans_mice_no_context → Score 26.95
```